

VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY



SOFTWARE ENGINEENG FINAL REPORT

COMPUTER SHOP MANAGEMENT SOFTWARE

Instructor: **GV NGUYỄN NGỌC PHIÊN**

Executor: **NGUYỄN NGỌC BẢO UYÊN - 520H0440**

PHẠM NGUYỄN PHÁT ĐẠT - 520H0348

PHẠM THÀNH PHONG - 520H0399

LÊ NHẬT DUY - 520H0352

Course: **24**

HO CHI MINH CITY, 2022

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THANK YOU

First, we would like to express our sincere thanks and deep gratitude to Professor Nguyen Ngoc Phien, who has always supported and guided us wholeheartedly during the research and completion of the report. end of term.

Next, we would like to thank the Department of Information Technology, Ton Duc Thang University for creating conditions for me to study and research this subject. The faculty has always been ready to share useful knowledge as well as to share experiences of referencing documents, which helps not only in carrying out and completing research projects but also in learning and training. during practice at Ton Duc Thang University in general.

Finally, after a period of studying in class, we completed the research project thanks to the guidance, help and knowledge learned from the teachers. Due to the limitation of knowledge and reasoning ability, the group still has many shortcomings and limitations. We would like to ask for your guidance and contributions to make our research more complete. Moreover, thanks to the suggestions from teachers and friends, we will complete better in future research papers. We hope that all of our teachers and friends - who always care about and support me - are always full of health and peace.

WE THANK YOU!

PROJECT COMPLETED AT TON DUC THANG UNIVERSITY

I hereby declare that this is my own project and is guided by Mr Nguyen Ngoc Phien. The research contents and results in this topic are honest and have not been published in any form before. The data in the tables for analysis, comments and evaluation are collected by the author himself from different sources, clearly stated in the reference section.

In addition, the project also uses a number of comments, assessments as well as data of other authors, other agencies and organizations, with citations and source annotations.

If I find any fraud, I will take full responsibility for the content of my project. Ton Duc Thang University is not related to copyright and copyright violations caused by me during the implementation process (if any).

Ho Chi Minh City, day month year

TEACHER'S CONFIRMATION AND ASSESSMENT SECTION

The confirmation part of the instructor

Ho Chi Minh City, day month year

Author

(Sign and write full name)

The evaluation part of the teacher marks the exam

Ho Chi Minh City, day month year

Author

(Sign and write full name)

SUMMARY

The report is made using the knowledge learned and materials provided by the lecturer, in addition to the reference of documents at Ton Duc Thang University Library.

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CHAPTER I: INTRODUCTION

Topic name

Computer shop management software.

1. Reasons to choose a topic

Nowadays, the management work in agencies in general and the management of computer stores in particular has been applied on computers because of the huge number. If we use the method of storing documents or documents, it will be very difficult to manage such as when we want to find information about an item or want to change the information of an item, it will be very difficult. with a huge variety of stores. From those difficulties, management software has been researched and developed to support management. The application of information technology to management not only reduces risks and errors but also helps managers feel comfortable at work.

For these reasons, our group decided to choose the topic "Computer shop management software" to contribute to solving the problem for managers to update information quickly, look up the processed data accurately in all aspects. order, price and revenue for subsequent operations or functions, easy to use with user-friendly graphical interfaces.

2. Requirement engineering

a) Require function

Admin: Customer consultation, order management, product management, user information list management

b) Non-functional requirements

i. Performance requirements

- Data is updated quickly, with complete consistency.

- Information received quickly and effectively through the website.
- Friendly interface, easy to use, fast response speed (under 1s for 1000 requests).
- Capable of serving a large amount of traffic at the same time.

ii. Safety requirements

- Capable of storing state, backing up the system and restoring data when having problems (network attack, connection failure, hardware failure,...).

iii. Confidentiality

- Strict decentralization: Manage user rights through functions (screens) and data. Independently designed View, Add, Edit, Delete functions give customers more flexibility in organizing multiple users and controlling data.

iv. Software quality attributes

- Built on web platform and compatible with many browsers (Chrome, Opera, Firefox, mobile browsers,...) to use anytime, anywhere..
- Easy to maintain, upgrade, develop software after deployed. Interoperability, maintainability, portability, reliability, reusability, robustness, testability.
- The ability to reuse the software for a number of other information management purposes is very powerful.
- Absolute data control, safe and secure.

3. Business process

Customers buy products online (pay via domestic bank card or e-wallet) or pay on delivery. Customers can view and search for products even without logging in, but if customers want to buy products, customers will be required to have an account and to get an account, users will register. To

make a purchase, the user must add the product to the cart and pay online with a domestic bank card or via an e-wallet, after the payment is completed, the system will receive and process the order, the customer You can view the progress of your order through the order view function.

4. Fact survey

Interviewer	Interviewees	Interviewee's role	Interview questions	Answer
Lê Nhật Duy	Nguyễn Lương Phúc	User	Hello, in your opinion, what is the most outstanding thing about an online computer selling site?	Hi, according to years of experience in buying computers online, the most outstanding thing is the convenience and speed of payment and receipt. What stores can't provide
Lê Nhật Duy	Nguyễn Công Hậu	User	Hello, can you tell me what are the disadvantages of these online computer selling sites?	Hi, as far as I know, along with my experience of using one or two sites to sell computers online, one of the biggest disadvantages of these sites is the ability to connect to the Internet. If one day you don't have 4G or Wifi and you

				want to buy an item online, it's impossible
Nguyễn Ngọc Bảo Uyên	Lê Hồ Bảo Khang	Developer	Hi, can you tell me what is difficult when launching an online computer selling website?	As I see it, it is the risk of data loss. Because all business data is stored at the company's computer server along with hundreds of other customers, the virus can easily attack and crash the whole system. At this point, businesses will almost lose all data. All information about customers, revenues and expenditures is difficult to recover this. The consequences are hard to imagine.
Phạm Nguyễn Phát Đạt	Trần Tường Minh	Admin	Hello, I want to know what is one of the advantages of selling	Hi, an interesting question, one of the advantages is that the initial implementation cost

			computers online?	is quite low. This can be seen as the short-term advantage of this job. When starting a business, Only 2-3 million / 1 year / paid for the first time. Every year, it is necessary to collect additional money to maintain the server, the staff to ensure your data will be secure. In fact, the 1-year cost of an online computer is only slightly cheaper than the perpetual cost of an offline computer.
Phạm Thành Phong	Nguyễn Minh Trí	User	Hello, can you share with me the benefits of buying computer online that you know?	Hi, in my opinion, the benefits of buying computer online are the ability to respond and make quick payments, a team of enthusiastic consultants, and

				support for many payment methods.
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Table 1. Actual interview table

CHAPTER II: SPECIFIC DESCRIPTION

1. System specification

The system "Computer shop management software" includes the objects Customer and Admin.

On the Customer side: When you want to buy goods at the website, you must have an account, customers can have an account by registering an account before. Customer can view products and add to cart and pay for his products by paying via card or ship-cod, then customer can review his order.

On the Admin side: Admin will manage suppliers, product catalogs, accounts, personal data of users. After the user places an order and payment, the Admin will have the task of receiving and processing the order, after processing is completed, the Admin will continue to handle the delivery to the customer.

2. Target audience

The system is aimed at computer sales managers who want to expand their business and manage their time and effort at work and optimize business profits.

3. Range

The system ensures high security, wide application, language support for users, suitable for multi-dimensional business, widely operating in the country.

4. Possibility

Technically, the programmer has many ways to optimize for this system, so this project can be successful.

In terms of finance, this system is aimed at owners and businessmen who want to expand the business market and attract domestic investment.

5. Actors in the system

Ordinal number	Actor	Describe
1	Admin	As a sales system manager, member customer list manager, product list manager.
2	Customer	As regular customers, visit the website but do not register for an account.
3	Member Customer	As customers who have registered an account, can buy goods on the website.
4	Payment portal	Payment methods.

Table 2. Agent table in the system

6. Use Cases in the system

Ordinal number	Use case Name	Describe
UC01	Sign Up, Sign In	Customers will login if they already have an account or register if they don't have an account.
UC02	Management of personal information	Customers use to view and edit personal information
UC03	Use case Cart management, Buy Items	Customers can view product information, add products to cart and can checkout products or remove products from cart.

UC04	Use case Payment	Customers can choose to pay via 3 methods, pay on delivery, pay via bank card, pay via e-wallet
UC05	Use case View products	Customers can see details of a product such as price, supplier
UC06	Use case Search for products	Customers can search for products by product name or supplier
UC07	Use case View orders	Customers can view the status of their orders and review completed orders.
UC08	Use case Customer support	Customers can be supported through 2 ways: online chat or call via hotline
UC09	Use case Manage member lists	Admin will be able to manage adding, deleting and editing customer information.
UC10	Use case Product portfolio management	Admin can manage products by adding, deleting or editing product information.
UC11	Use case Receiving and processing orders	Admin will be able to receive order processing, view order information and handle shipping it.

Table 3. Table of use cases in the system

7. System functions

Ordinal number	Function	Describe
1	Register	Allow customers to register for a member account.
2	Login	Allow admins and members to log in to the system.

3	Personal information management	Allow customers who already have a member account, admin to view personal information and edit personal information.
4	Cart	Allow customers to add and remove products in the cart.
5	Pay	Allow customers to pay bills with 3 methods.
6	Search	Allow customers to search for products by name or supplier.
7	View product information	Allows customers to view product details, view products by vendor name, by price and products on sale
8	Product Management	Admin function, allowing admin to add products, delete products or edit product information.
9	View orders	Allows customers to view the status of current orders, or review all completed orders.
10	Order processing	Admin function, allowing admin to process pending orders and then ship.
11	Supplier Management	Allows admin to track, control and categorize product suppliers.
12	Manage member list	Allow admin to add, delete, change information of members in the list.
13	Customer support	Customers can contact support with the system through 2 methods: online chat and hotline.

Table 4. Table of functions in the system

CHAPTER III: DETAILED DESIGN

1. General use case diagram

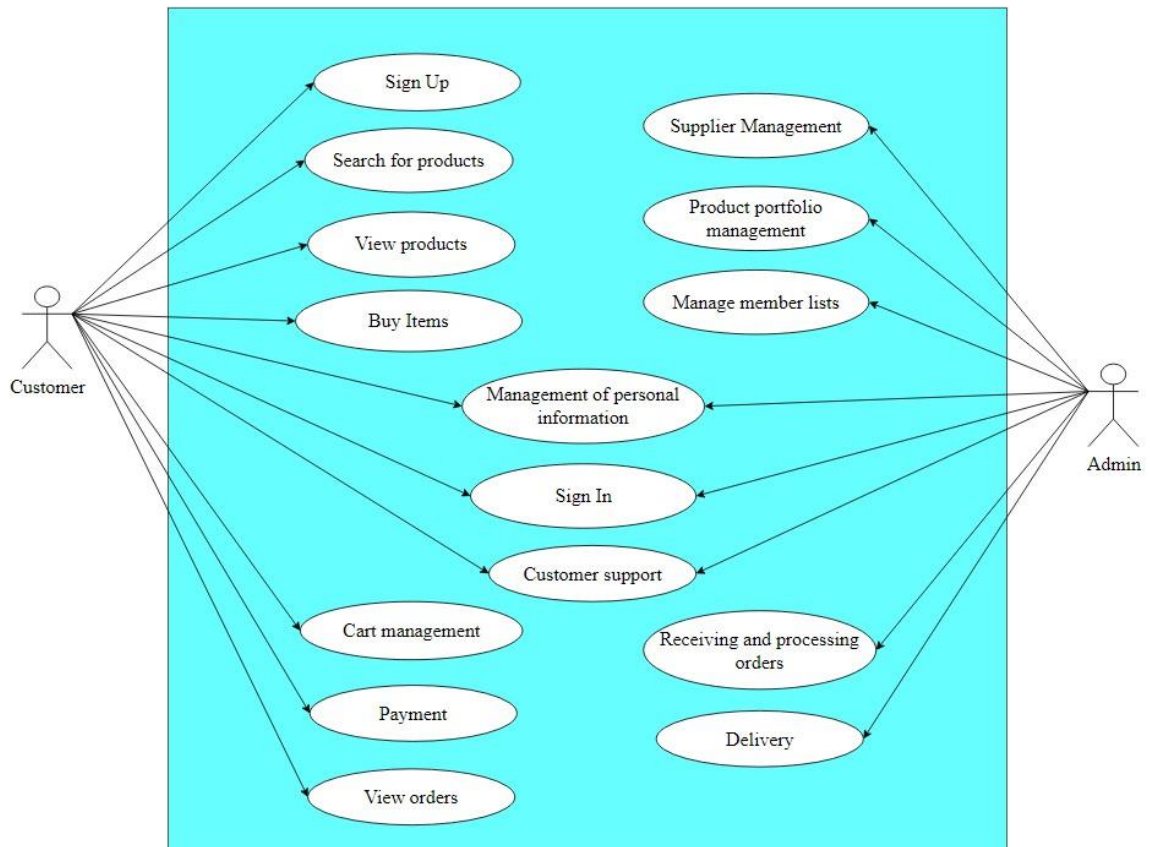


Image 1. General use case diagram

1. Use case diagram

i. Use case Sign Up, Sign In

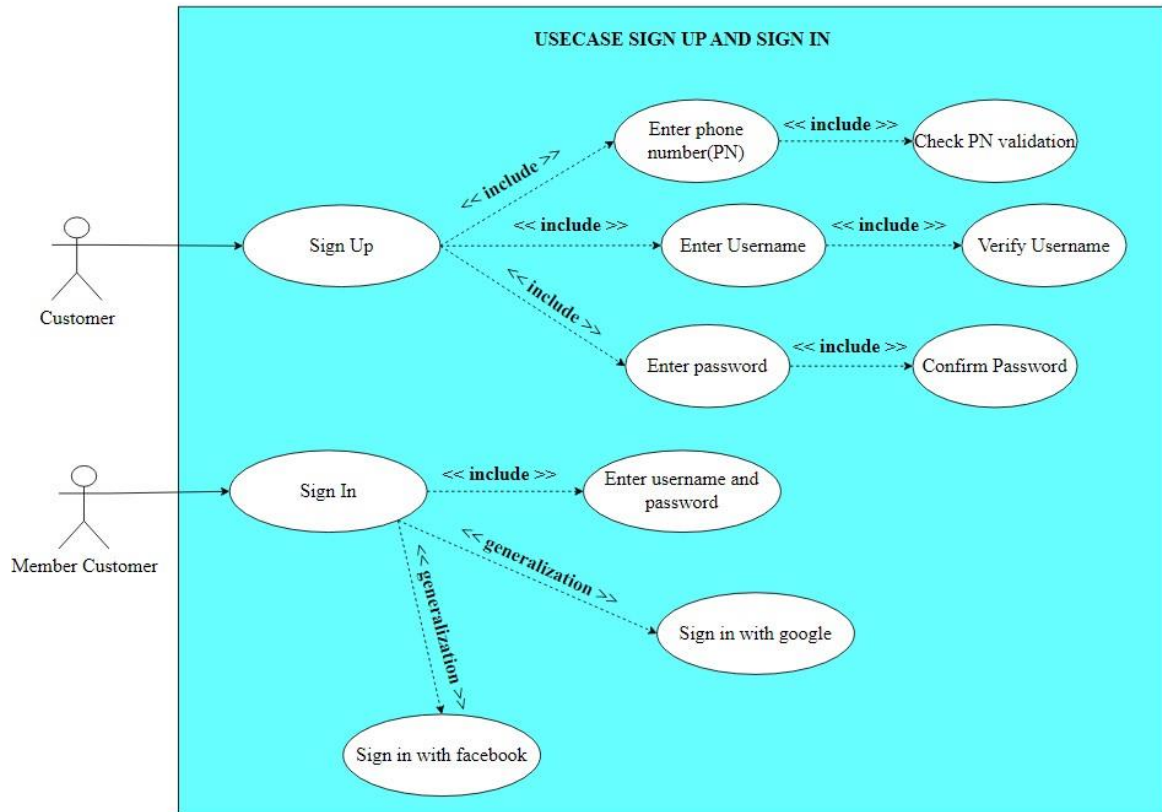


Image 2. Use case Sign up, Sign In

ii. Use case Management of personal information

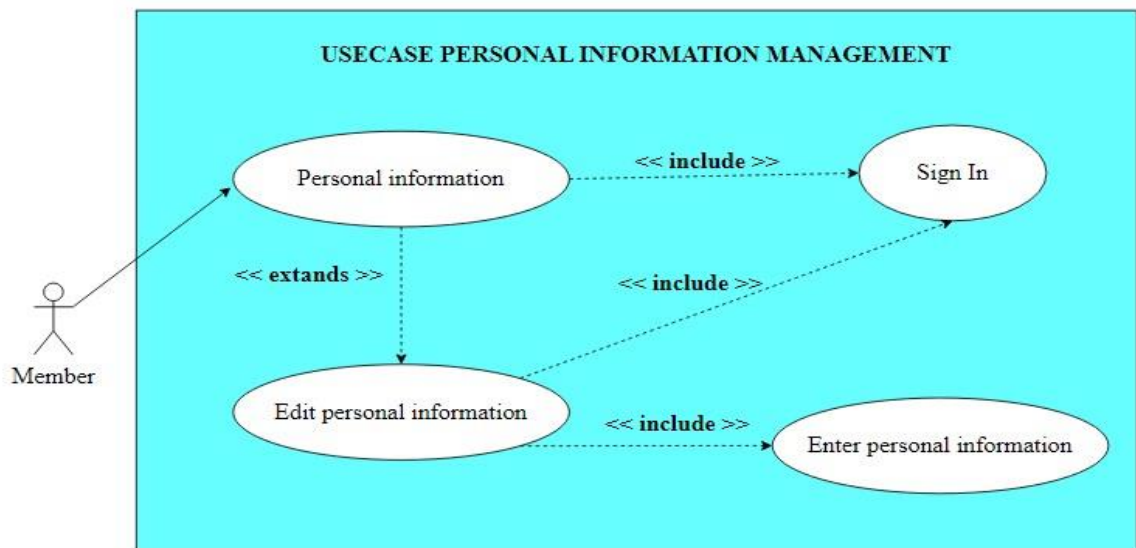


Image 3. Use case Management of personal information

iii. Use case Cart management, Buy Items

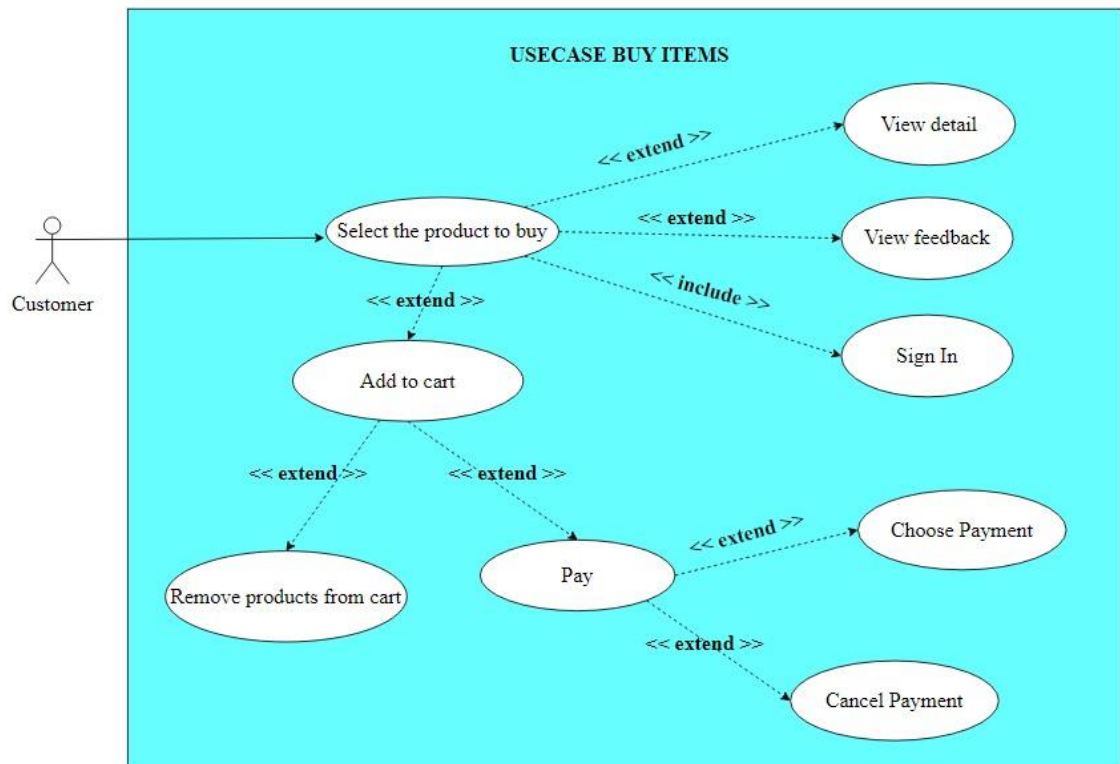


Image 4. Use case Cart management, Buy Items

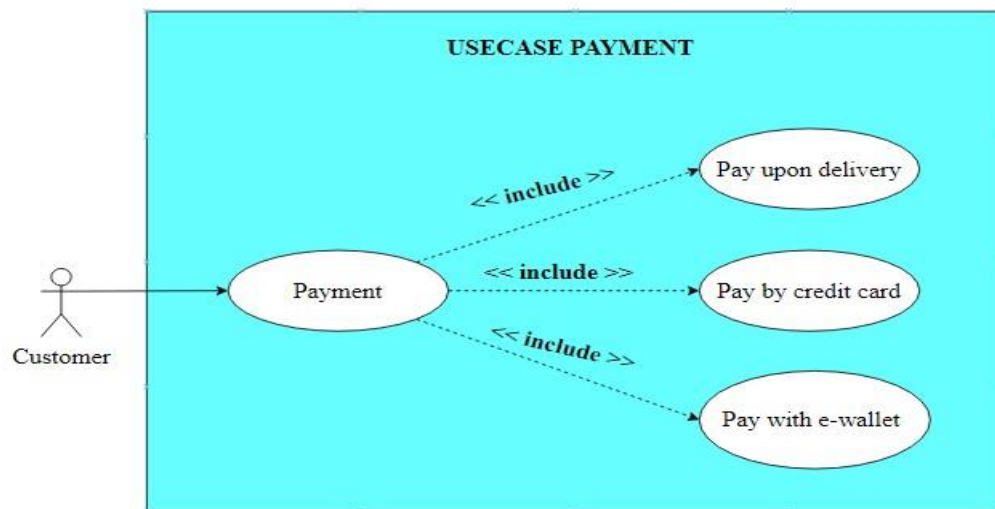
iv. Use case Payment

Image 5. Use case Payment

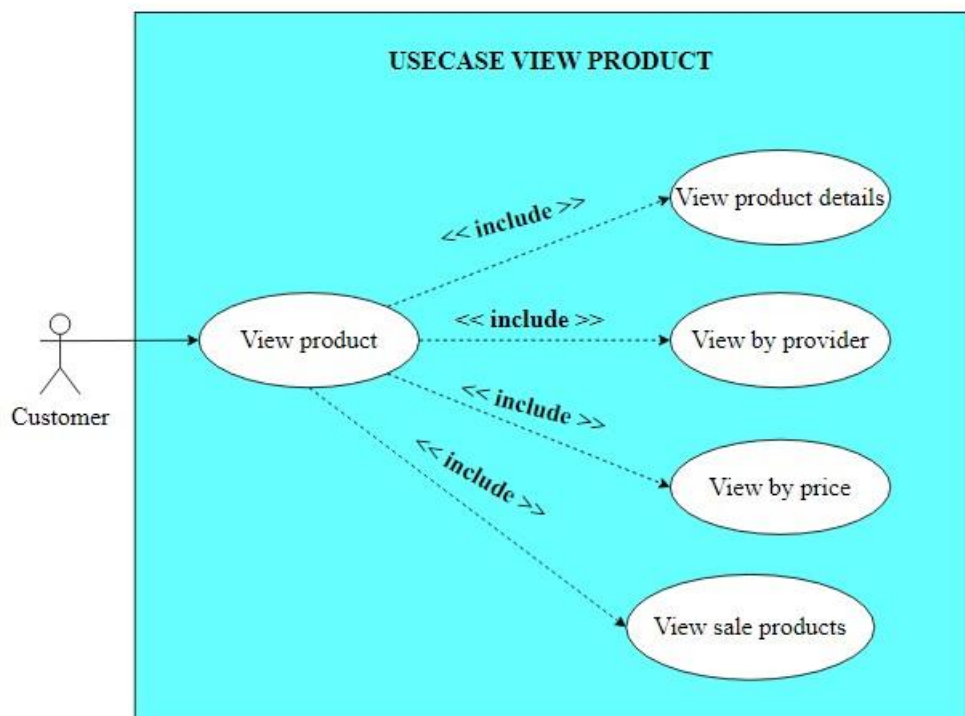
v. Use case View products

Image 6. Use case View product

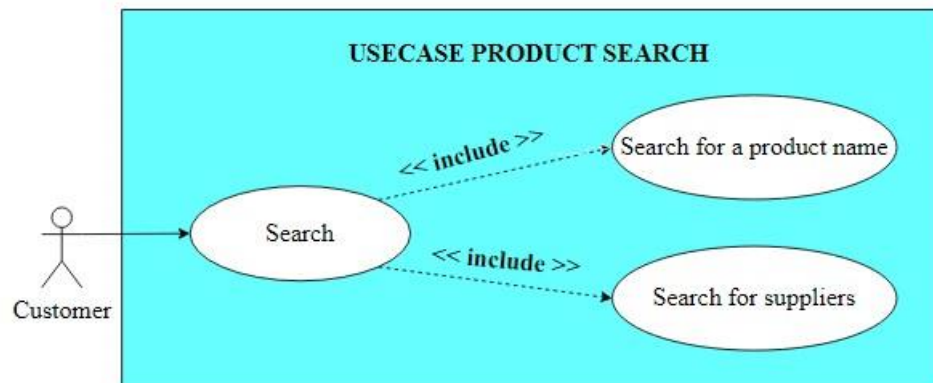
vi. Use case Search for products

Image 7. Use case Search for products

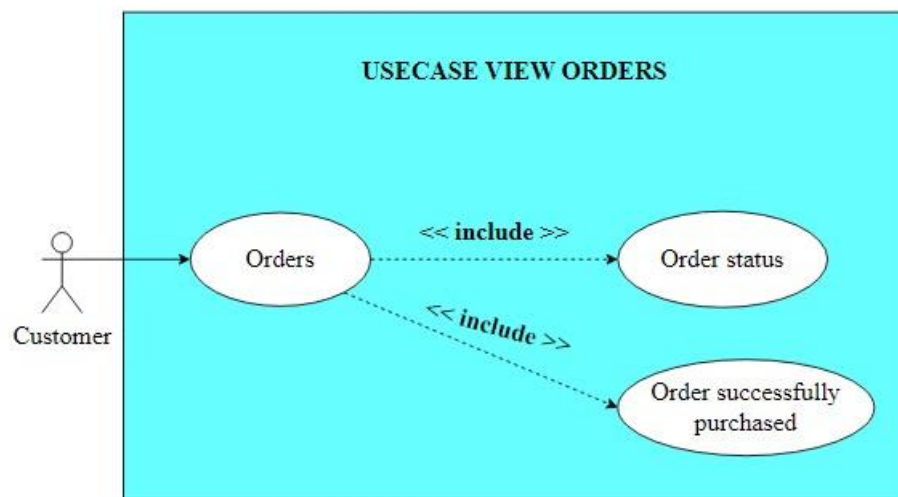
vii. Use case View orders

Image 8. Use case view orders

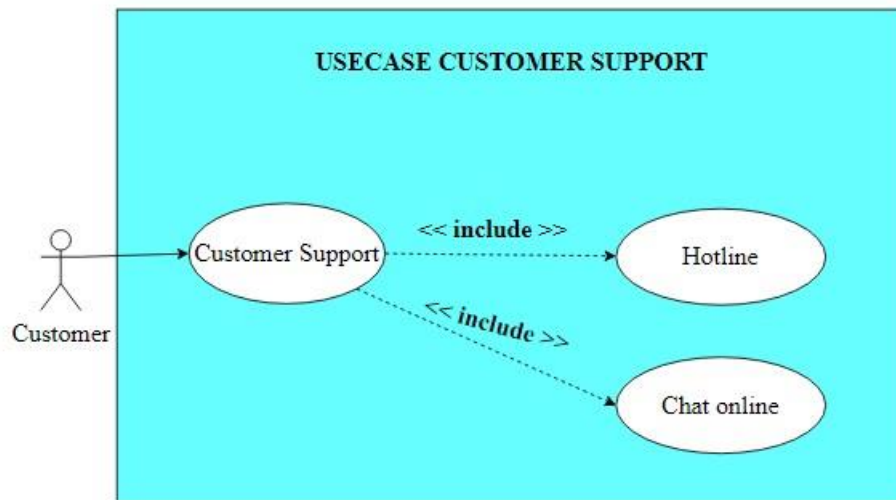
viii. Use case Customer support

Image 9. Use case Customer support

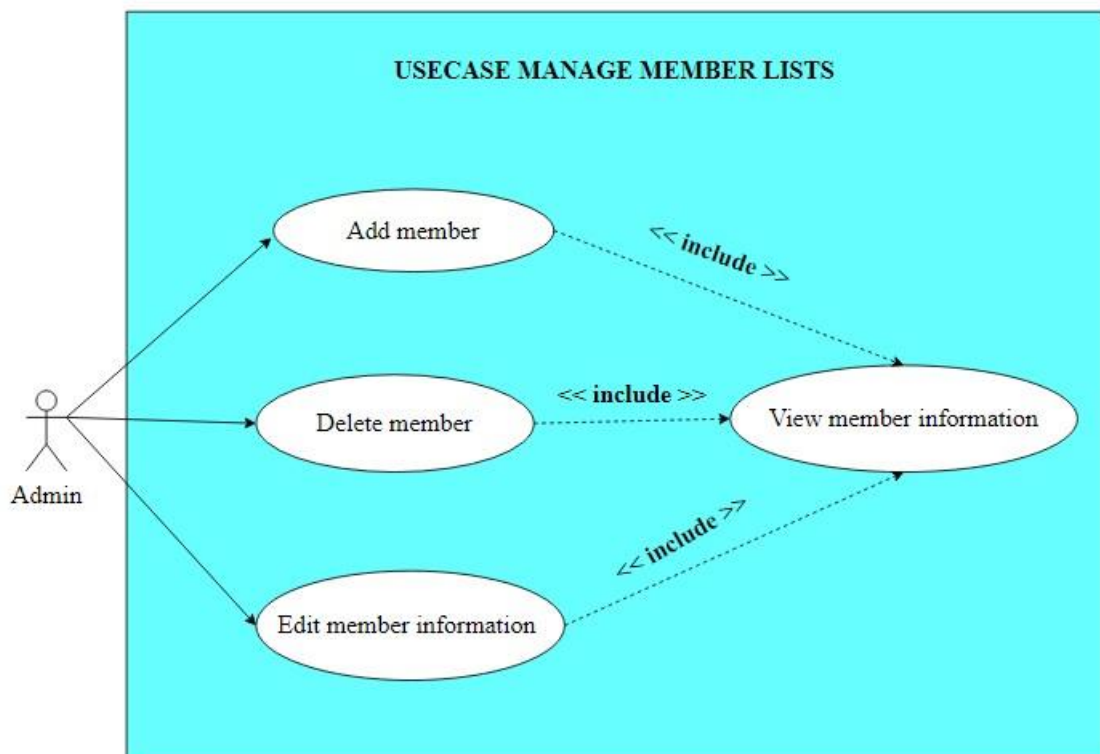
ix. Use case Manage member lists

Image 10. Use case manage member lists

x. Use case Product portfolio management

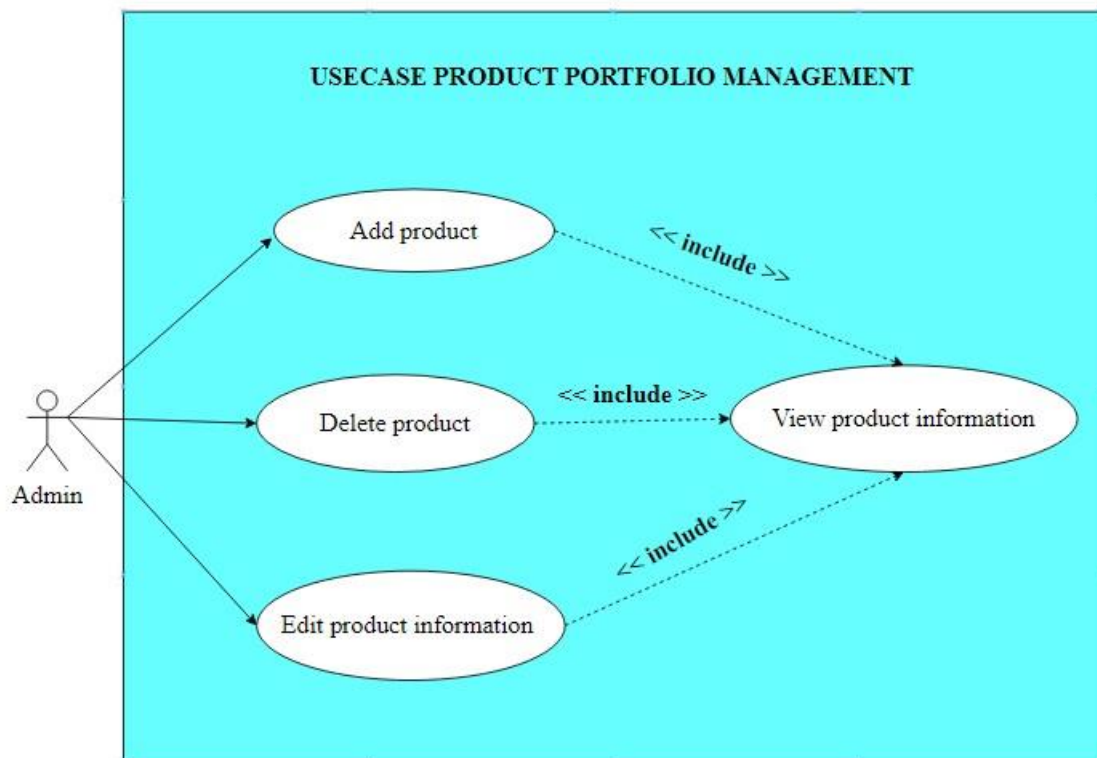


Image 11. Use case Product portfolio management

xi. Use case Receiving and processing orders

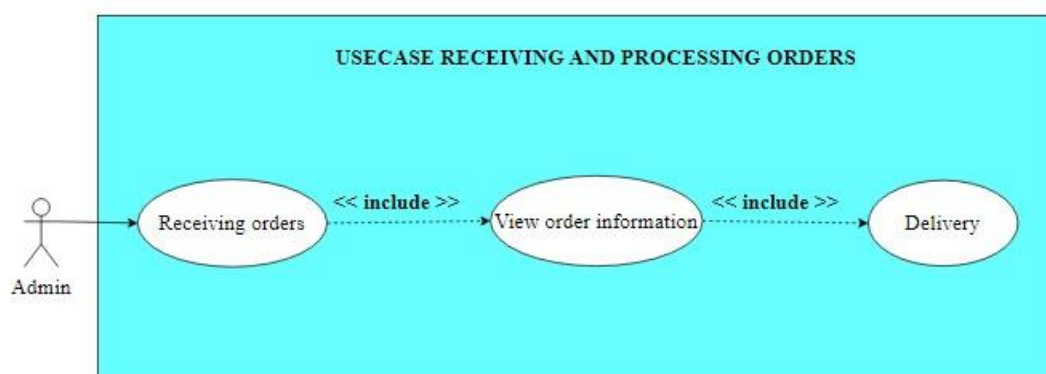


Image 12. Use case Receiving and processing orders

2. Use case specification

i. Use case Sign Up, Sign In

Use Case Name	Sign up/Sign in
Scenario	Users want to use the website to make purchases
Triggering event	Users who want to have an account and use the website
Brief description	Allow users to log in to the site to make purchases
Actors	Customer, Member Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the site _ User need to have internet
Post conditions	_ User successfully registered _ User successfully logged in
Flow of activities(Actor)	1. User selects "Register". 1.2. Enter your phone number. 1.3. Enter password. 1.4. Enter account 1.5. Click the "sign up" button. 2. User selects "Login" 2.1 Login with account and password 2.2 Sign in with google 2.3 Login with Facebook
Flow of activities(System)	1.1 Display the "Register" interface 1.5.1 Show successful registration. 2.1 Display the "Login" interface. 2.3.1 Display successful login and go to home page.

Exception conditions	1.2 If you enter an incorrect or invalid phone number, you will be asked to re-enter it. 1.3 If you confirm the password is incorrect you will be asked to re-enter it 2.1 If you enter the wrong account or password, you will be asked to re-enter it.
----------------------	---

Table 5. Specification sheet use case sign up, sign in

ii. Use case Management of personal information

Use Case Name	Personal information management
Scenario	When users want to edit personal information
Triggering event	Users use the website to edit personal information
Brief description	Allow users to change, edit their personal information
Actors	Member
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User must have internet connection
Post conditions	_ User successfully logged in
Flow of activities(Actor)	1. User selects "Personal information" 1.2 User selects "Edit personal information" 1.4 The user enters the information to be edited.
Flow of activities(System)	1.1 Display interface "Personal information" 1.3 Display information editing interface

	1.5 After the user has finished editing, the system will update and display the correct information with the information edited by the user.
Exception conditions	1.2 If the user is not logged in, the system will redirect the user to the login page

Table 6. Specification sheet Use case Management of personal information

iii. Use case Cart management, Buy Items

Use Case Name	Buy Items
Scenario	When a user makes a purchase
Triggering event	Users use the website to view products and make purchases
Brief description	Allows users to view product information, add and remove products to the cart, and pay for products in the cart.
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User must have internet connection
Post conditions	_ User successfully logged in _ User must select product to add to cart
Flow of activities(Actor)	1. User selects the product to buy 1.3 Users choose to view product information or add to cart. 1.5 In the shopping cart interface, the user chooses to pay for the product or remove the product from the cart 1.7 If the user chooses to pay, the user will continue to choose the payment method or cancel the payment

Flow of activities(System)	1.2 Display product interface will include product information and add to cart button. 1.4 Display cart interface 1.6 If the user chooses to remove the product, the cart interface will be updated with the removed product 1.6.1 If the user chooses to pay, then switch to the payment interface. 1.8 If the user chooses to cancel the payment, return to the shopping cart interface.
Exception conditions	1.3.1 To add products to the cart, if the user is not logged in, they will have to log in

Table 7. Specification sheet Use case Cart management, Buy Items

iv. Use case Payment

Use Case Name	Payment
Scenario	The user wants to pay for the product
Triggering event	Users want to have an account and use the website to pay for products
Brief description	Allows users to pay for products using 3 different methods
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User need to have internet
Post conditions	_ User must login successfully _ User must have the product to buy in the cart

	_ User must meet 1 of 3 payment methods
Flow of activities(Actor)	1. User chooses to pay 1.2 User chooses 1 of 3 payment methods 1.4 User confirms payment and waits for order to be processed
Flow of activities(System)	1.1 Show payment interface and show 3 payment options 1.3 Display payment interface according to user selection.
Exception conditions	1.4.1 The system will check if the account has enough money for payment methods by bank cards and e-wallets, if not enough, the order will not be processed.

Table 8. Specification sheet Use case Payment

v. Use case View products

Use Case Name	View Product
Scenario	User wants to see the product
Triggering event	Users want to have an account and use the website to view products
Brief description	Allow users to view products through different criteria
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User need to have internet
Post conditions	_ User must login successfully _ Users need to see the product through the criteria available on the system

Flow of activities(Actor)	1.User selects item “View product” 1.2 Users choose to view products through 1 of 4 available criteria
Flow of activities(System)	1.1 Display the interface “View product” 1.3 Display products according to the criteria selected by the user
Exception conditions	1.2.1 Users can only view products through available criteria

Table 9. Specification sheet Use case View products

vi. Search for products

Use Case Name	Product search
Scenario	Users want to find products
Triggering event	Users use the site to search for products
Brief description	Allows users to search for products by name and manufacturer
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User need to have internet
Post conditions	_ User must have product or supplier name
Flow of activities(Actor)	1.User selects item “Search” 1.2 User enters product or supplier name to search
Flow of activities(System)	1.1Display the search interface 1.3 Display products by name or supplier entered by user

Exception conditions	1.3.1 If there is no product information provided by the user, the system will report that it is not found.
----------------------	---

Table 10. Specification sheet Search for products

vii. View orders

Use Case Name	View orders
Scenario	User wants to see current order status and completed order
Triggering event	User has an account and wants to review completed orders or current order status
Brief description	Allows viewers to review completed orders or current order status
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User need to have internet
Post conditions	_ User must login successfully _ User must have a previous order
Flow of activities(Actor)	1.User selects “Orders” 1.2 User chooses to view current order status or review completed orders
Flow of activities(System)	1.1 Display the orders . interface 1.3 Display current order status interface if user selects this function 1.3.1 Display completed order review interface if user selects this function

Exception conditions	1.2.1 If the user has no previous order, these functions will have no data
----------------------	--

Table 11. Specification sheet use case View orders

viii. Customer support

Use Case Name	Customer support
Scenario	User wants to be supported by admin
Triggering event	User has an account and wants to be supported by admin
Brief description	Allows users to contact the admin for support
Actors	Customer
Related use cases	Must have
Stakeholders	Admin
Preconditions	_ User successfully entered the website _ User need to have internet
Post conditions	_ User must login successfully
Flow of activities(Actor)	1.User selects “Customer Support” 1.2 User chooses 1 of 2 ways to contact admin
Flow of activities(System)	1.1 Display the interface “Customer Support” with 2 contact ways for users to choose 1.3 If the user chooses the hotline, the user will be provided with the admin's phone number 1.3.1 If the user selects Chat online, the user will be redirected to the online messaging interface with the admin
Exception conditions	1.2.1 If the user uses the phone and selects the hotline function, the user will be redirected to the calling application on the phone and the number will be available.

Table 12. Specification sheet use case Customer support

ix. Manage member lists

Use Case Name	Manage member list
Scenario	Admin wants to manage the list of user information
Triggering event	Admin wants to add, delete and edit the list of user information
Brief description	Allow admin to add, delete, edit user information list
Actors	Admin
Related use cases	Must have
Stakeholders	Customer
Preconditions	_ Admin successfully entered the website _ Admin must have internet connection
Post conditions	_ Admin login admin account successfully
Flow of activities(Actor)	1.Admin selects the Add member function 1.2 Admin select the member to add and confirm 2.Admin chooses the function Delete member 2.2 Admin select the member to delete and confirm 3.Admin selects the Edit member information function 3.2 Admin selects member who needs to change information and then confirm
Flow of activities(System)	1.1 Display the Add member interface for Admin to edit 2.1 Display Delete member interface for Admin to edit 3.1 Display the Edit member information interface for the Admin to edit
Exception conditions	1.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed.

	2.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed. 3.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed.
--	---

Table 13. Specification sheet use case Manage member lists

x. Product portfolio management

Use Case Name	Product Portfolio management
Scenario	Admin wants to manage the list of product information
Triggering event	Admin wants to add, delete and edit product information list
Brief description	Allow admin to add, delete, edit product information list
Actors	Admin
Related use cases	Must have
Stakeholders	Customer
Preconditions	_ Admin successfully entered the website _ Admin must have internet connection
Post conditions	_ Admin login admin account successfully
Flow of activities(Actor)	1. Admin selects the Add product function 1.2 Admin select the member to add and confirm 2. Admin selects the function Delete product 2.2 Admin select the member to delete and confirm 3. Admin selects the Edit product information function 3.2 Admin selects the member who needs to change the information and confirm
Flow of activities(System)	1.1 Display Add product interface for Admin to edit 2.1 Display Delete product interface for Admin to edit

	3.1 Display Edit product information interface for Admin to edit
Exception conditions	1.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed. 2.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed. 3.2.1 If successful, the system will update and display again, if unsuccessful, the system will report update failed.

Table 14. Specification sheet Product portfolio management

xi. Receiving and processing orders

Use Case Name	Receiving and processing orders
Scenario	Admin receives and processes orders
Triggering event	Admin receives and reviews orders and processes them for delivery
Brief description	Allow admin to view order information, process for shipping
Actors	Admin
Related use cases	Must have
Stakeholders	Customer
Preconditions	_ Admin successfully entered the website _ Admin must have internet connection
Post conditions	_ Admin login admin account successfully
Flow of activities(Actor)	1. Admin selects "Receiving orders" 1.1Admin selects the "View order information" section to view the order details, if the order is valid or meets the

	requirements, it will continue to select the "Delivery" section. 1.4 Admin select "Delivery" and proceed with shipping processing
Flow of activities(System)	1.1 Display the interface of orders that need to be received and processed 1.3 Display the order details interface 1.5 Display the delivery processing interface
Exception conditions	1.2.1 If the order is approved but the stock is not enough, the system will notify the order is canceled.

Table 15. Specification sheet use case Product portfolio management

3. Activity diagram

i. Sign up

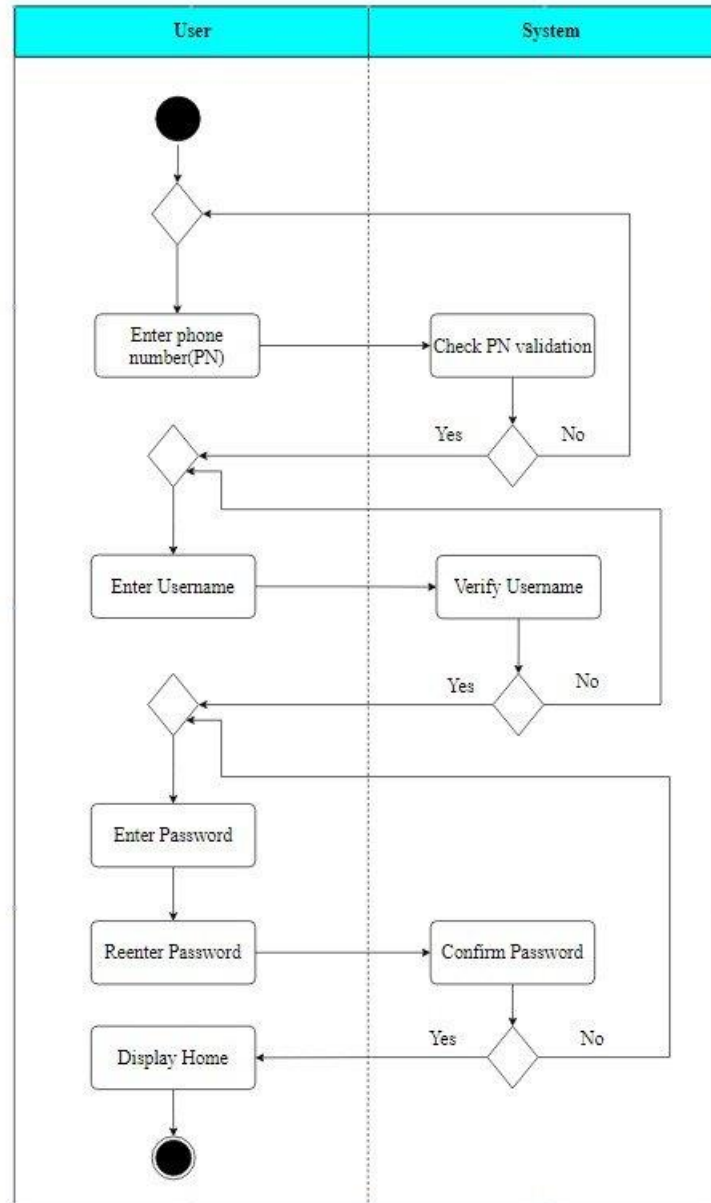


Image 13. Sign up activity diagram

ii. Sign in

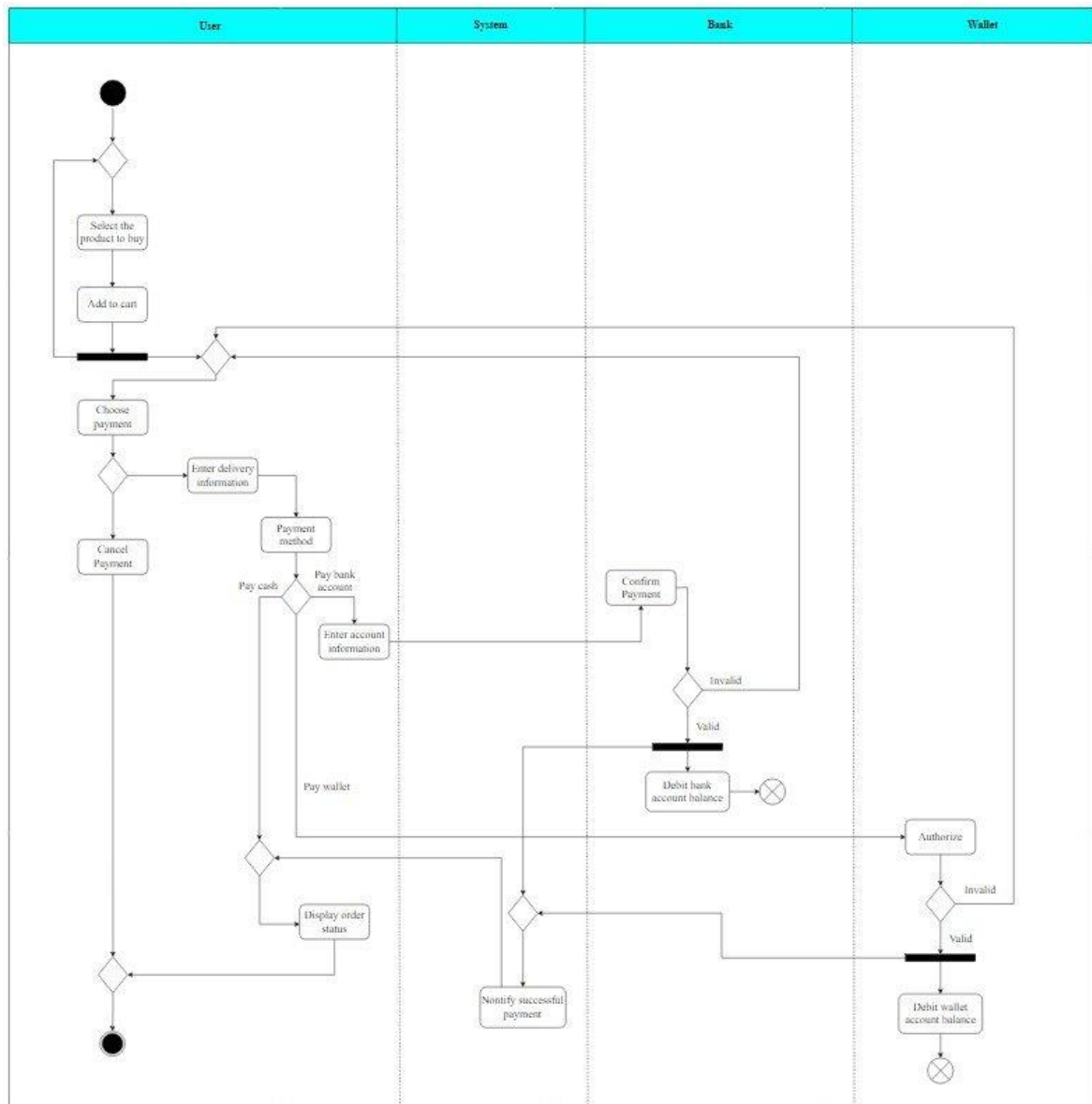


Image 14. Sign in activity diagram

iii. Management of personal information

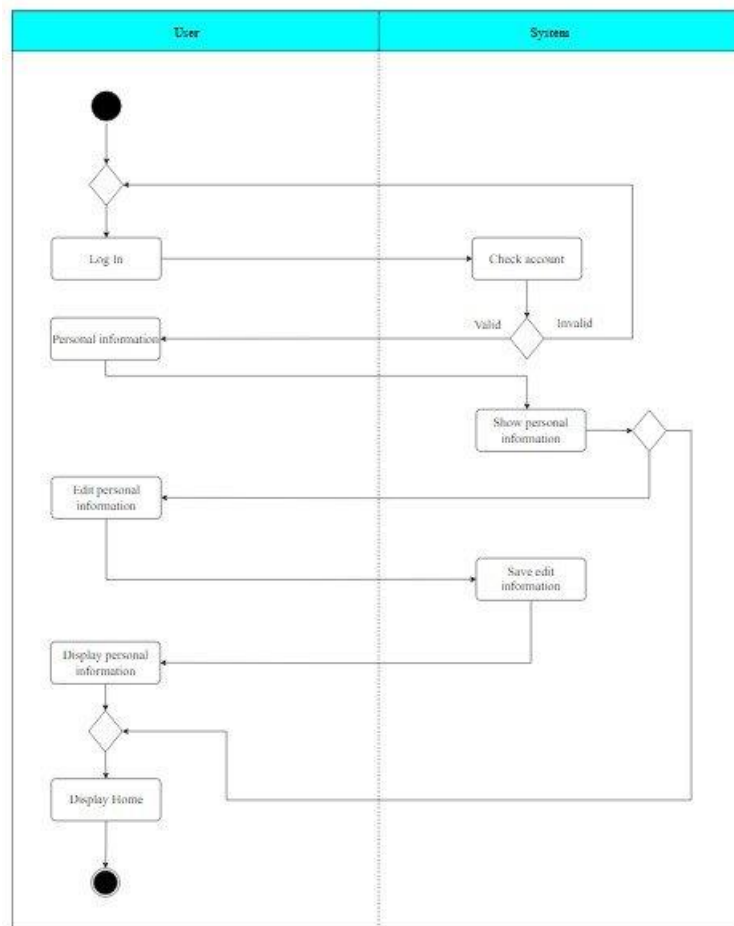


Image 15. Management of personal information activity diagram

iv. Payment and Buy Item

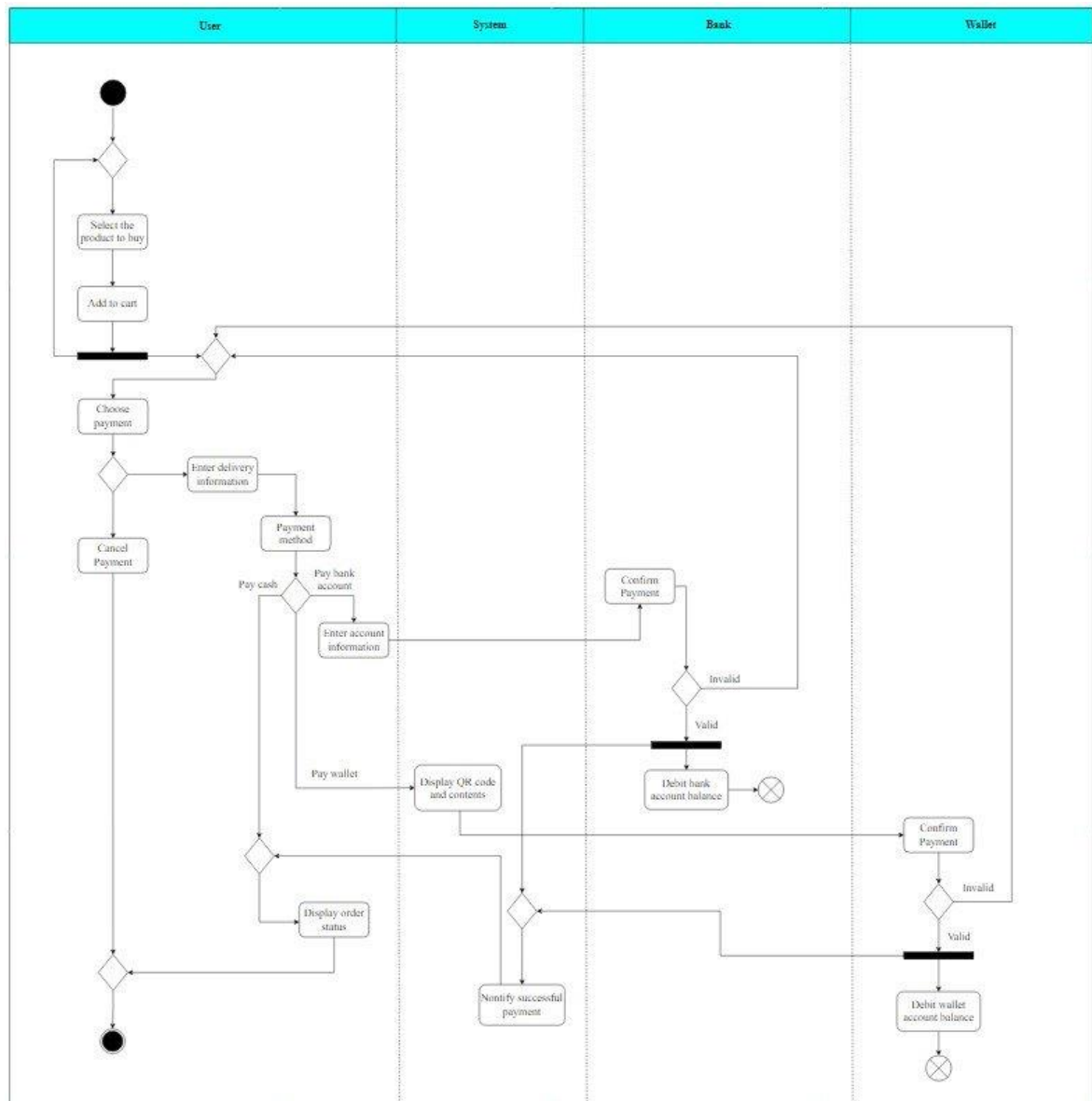


Image 16. Payment and Buy Item activity diagram

v. View products

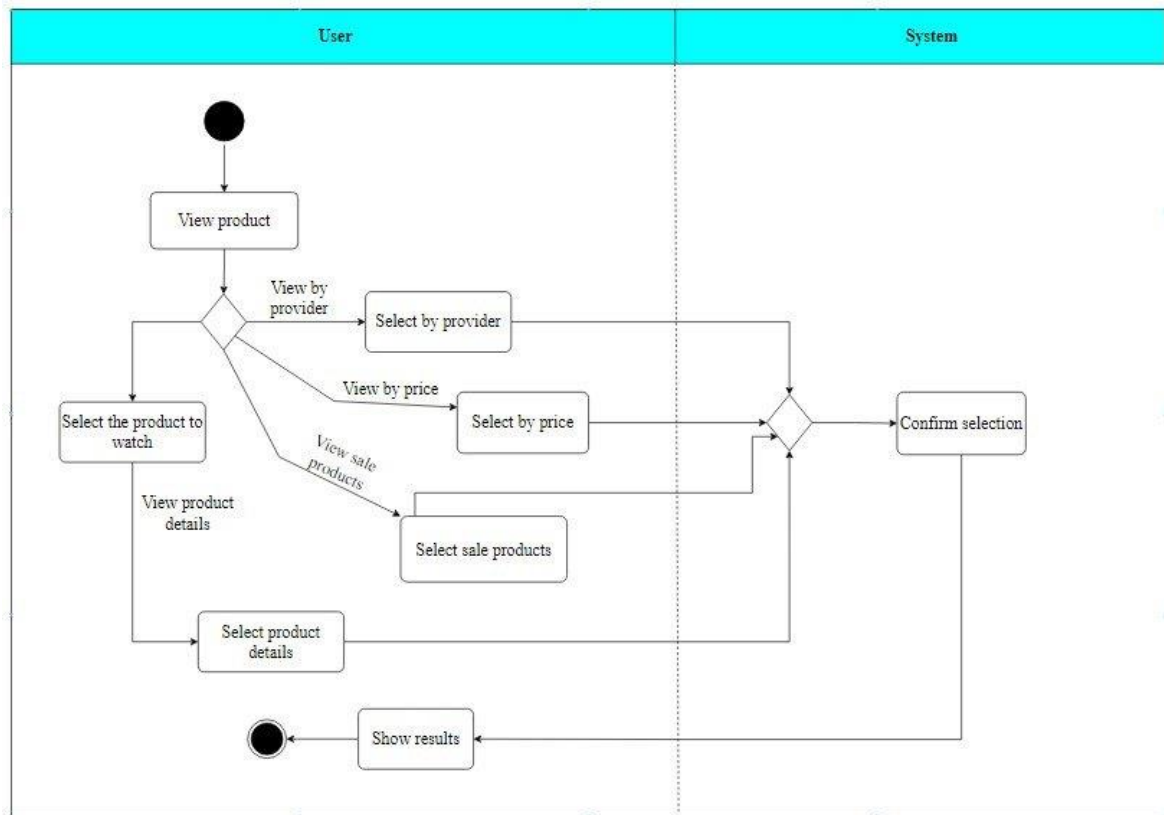


Image 17. View product activity diagram

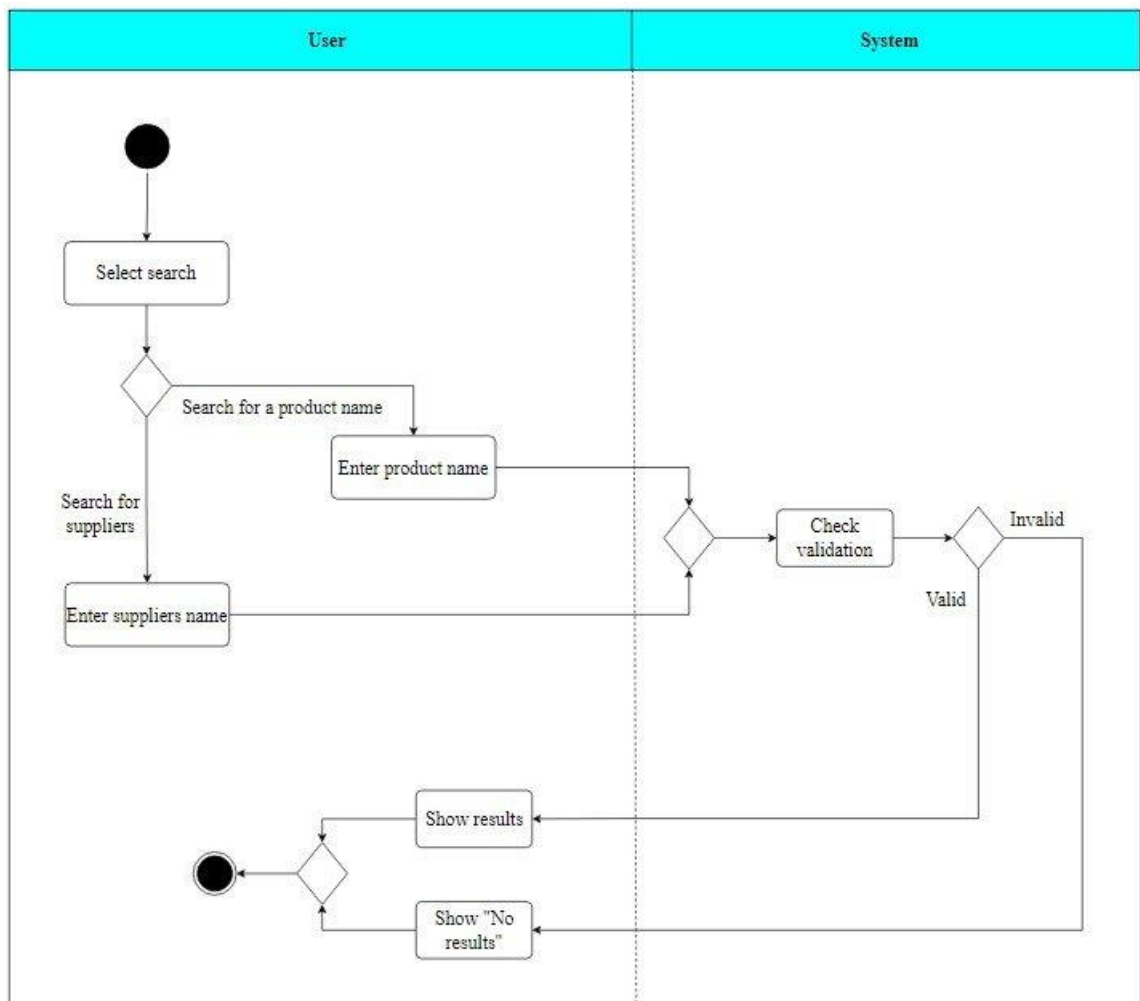
vi. Search for products

Image 18. Search for products activity diagram

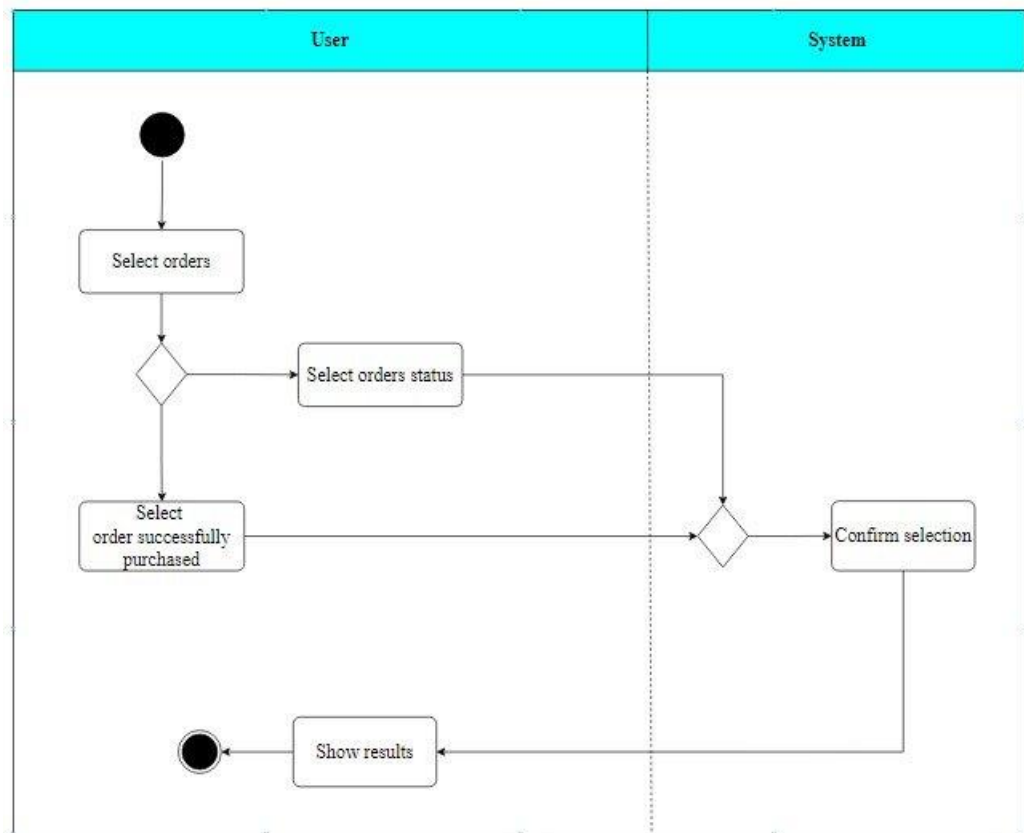
vii. View orders

Image 19. View orders activity diagram

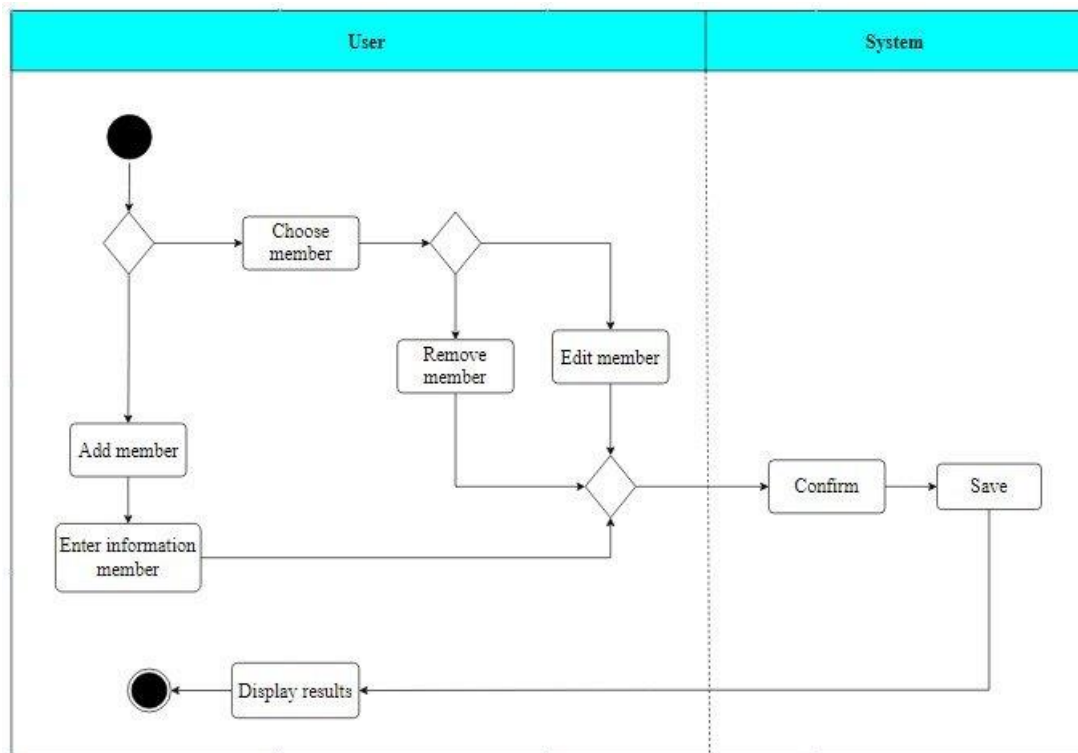
viii. Manage member lists

Image 20. Manage member lists activity diagram

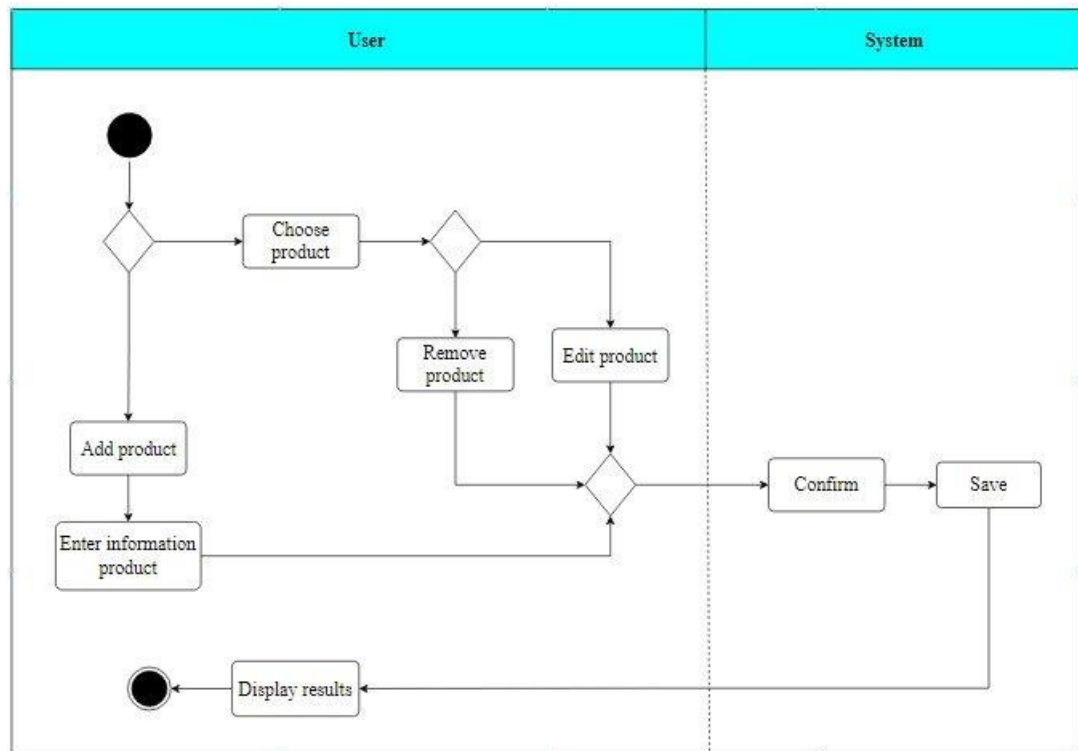
ix. Product portfolio management

Image 21. Product portfolio management activity diagram

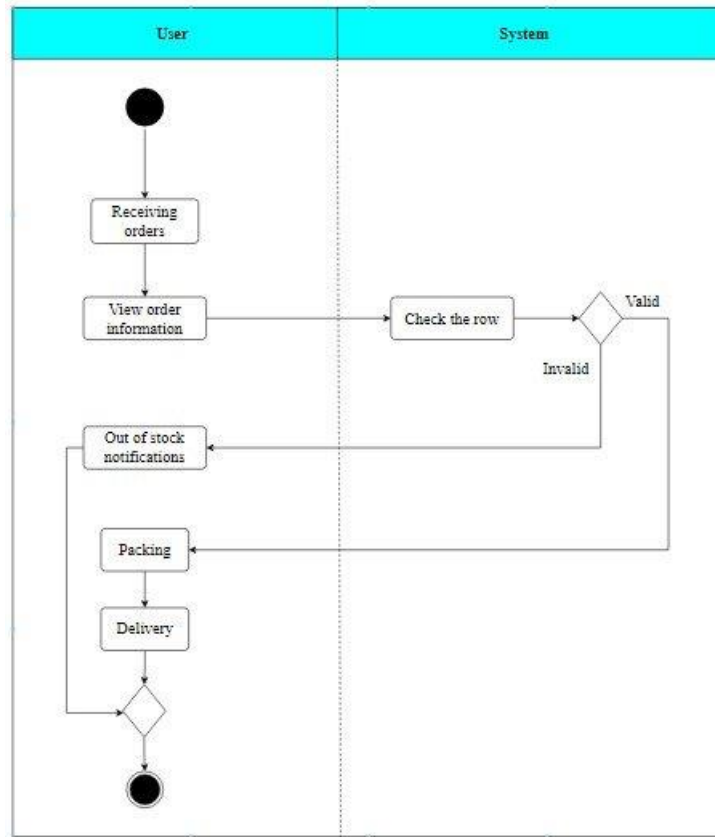
x. Receiving and processing orders

Image 22. Receiving and processing orders activity diagram

4. Sequence diagram

i. Sign up

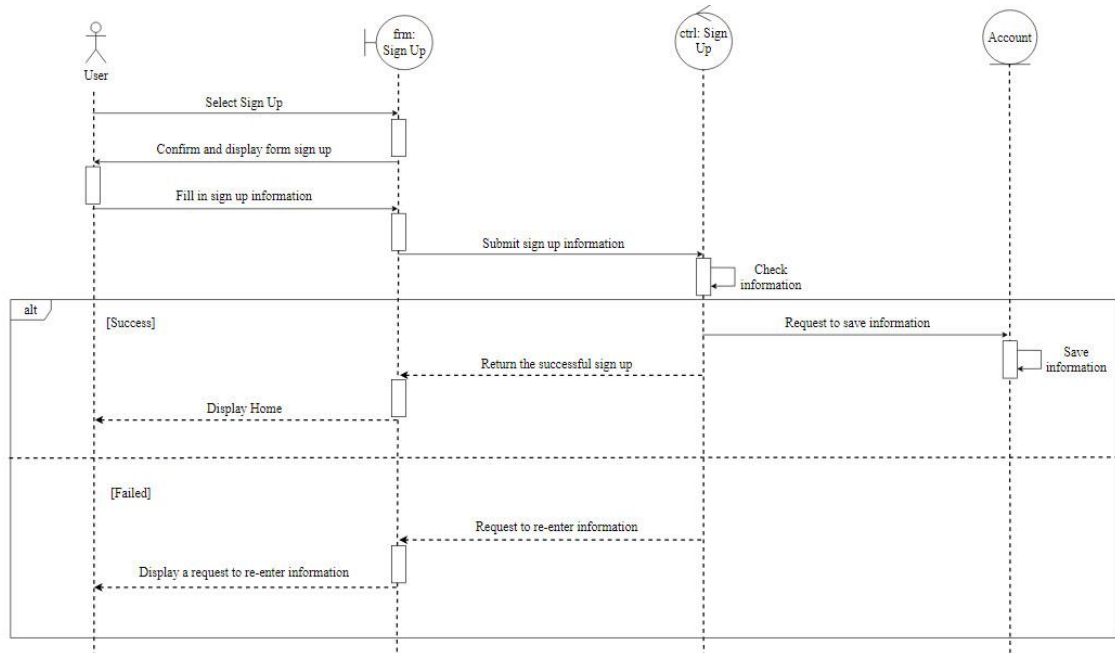


Image 23. Sign up sequence diagram

ii. Sign In

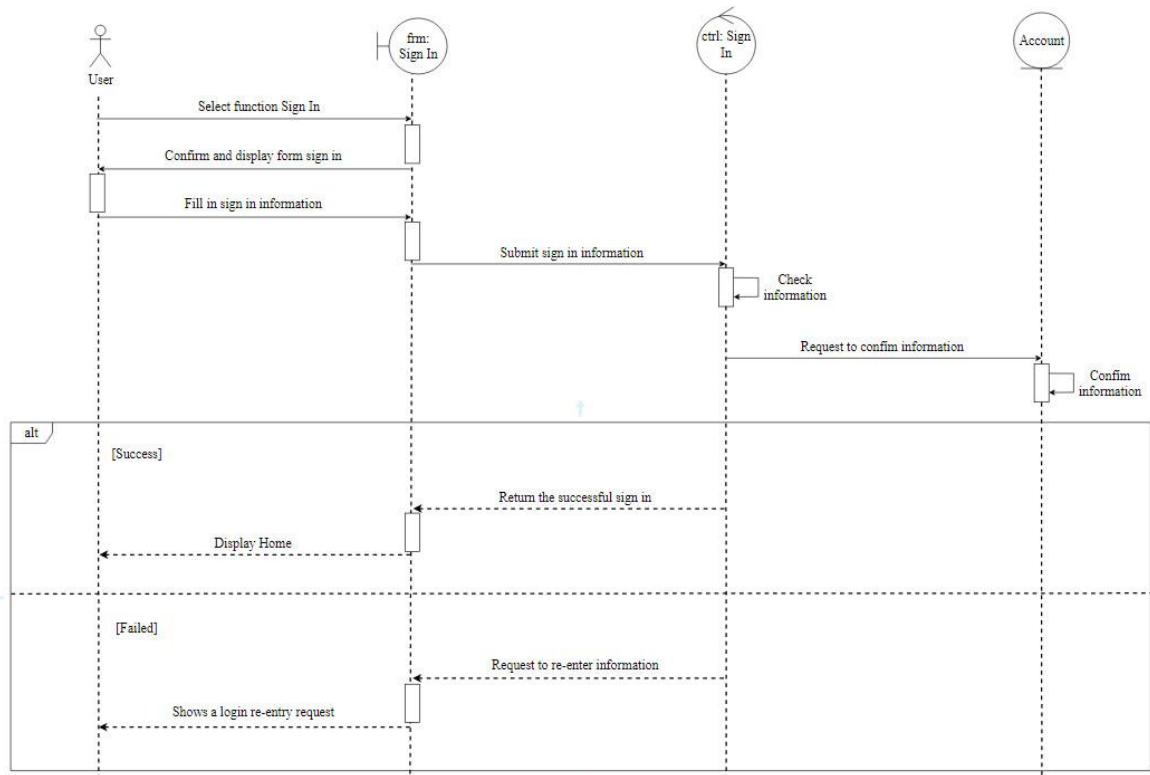


Image 24. Sign In sequence diagram

iii. Management of personal information

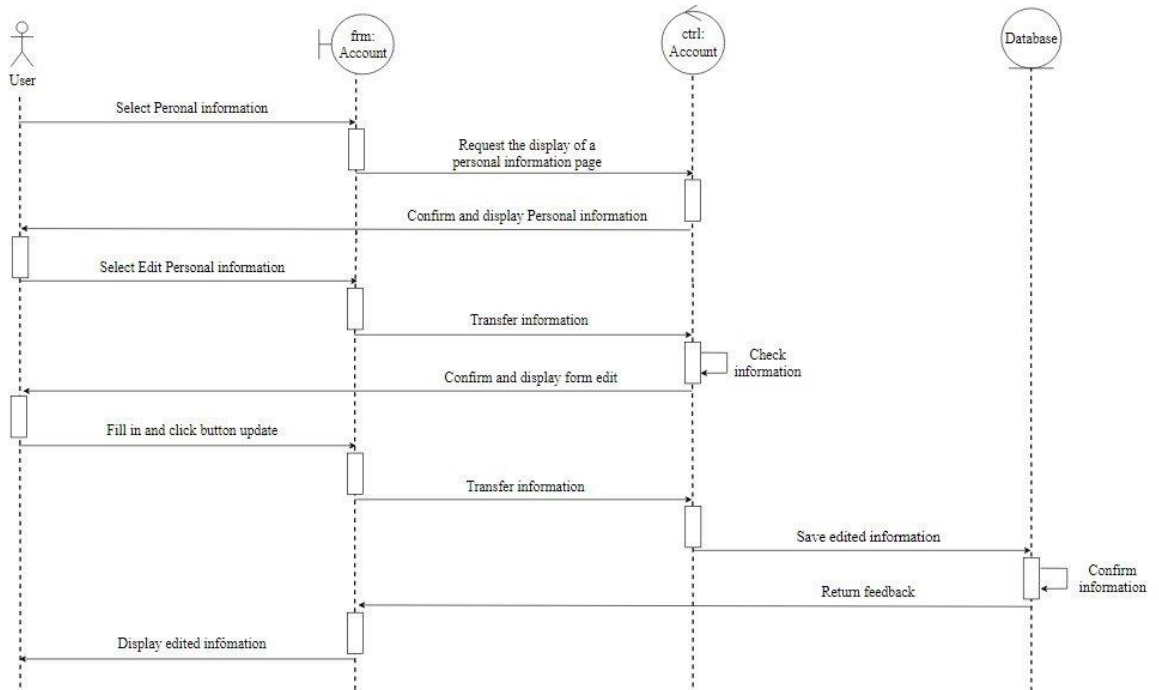


Image 25. Management of personal information sequence diagram

iv. Payment and Buy Item

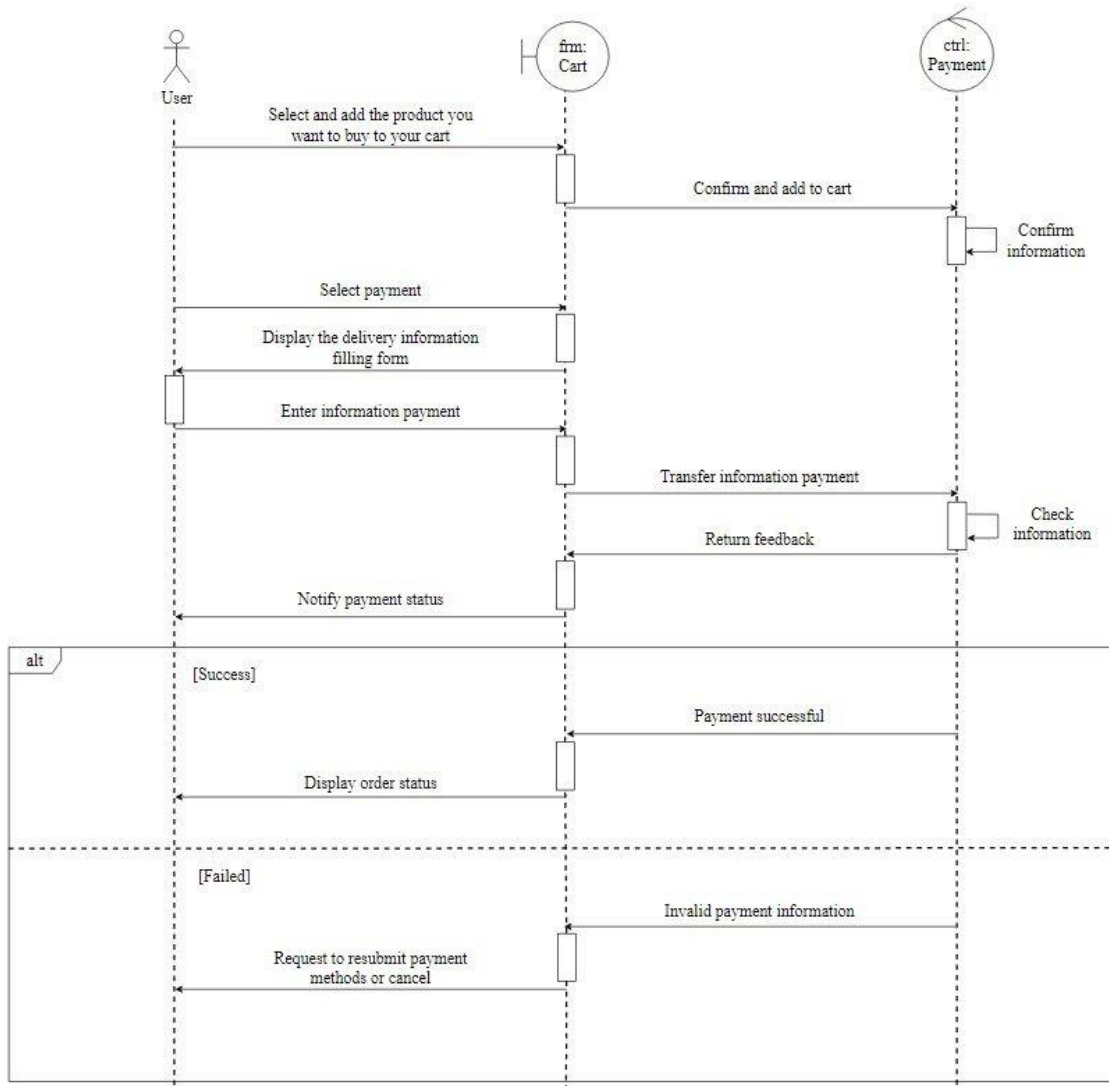


Image 26. Payment and Buy Item sequence diagram

v. View products

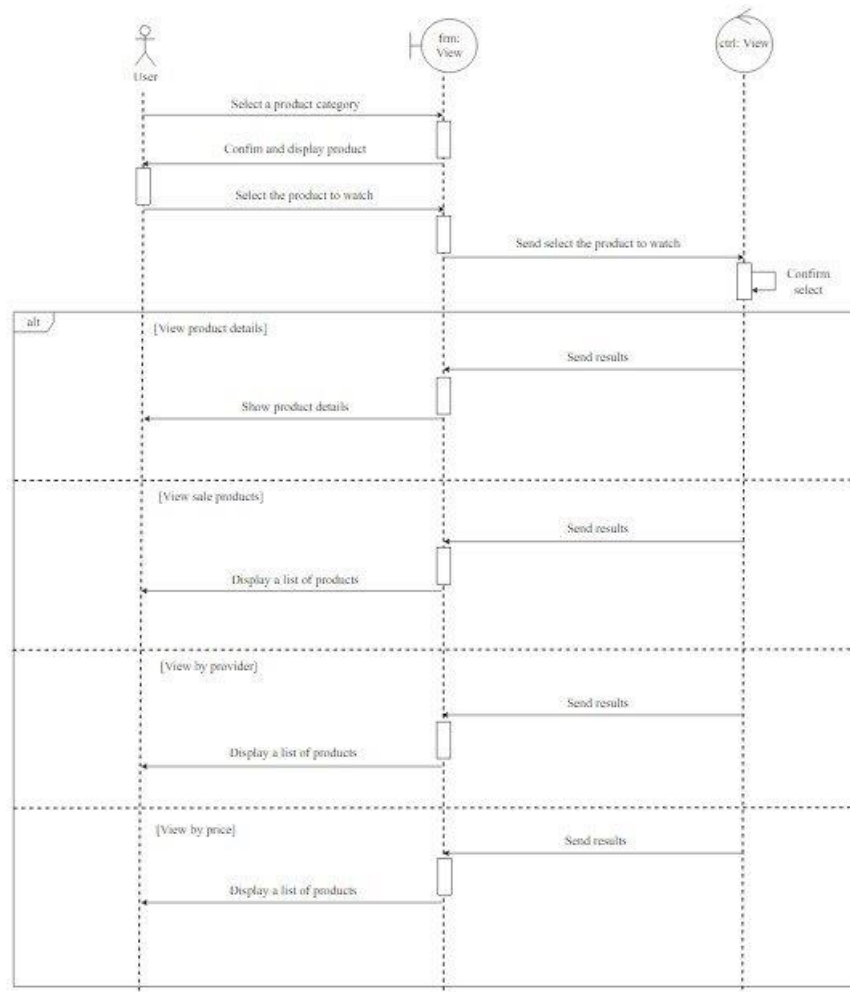


Image 27. View products sequence diagram

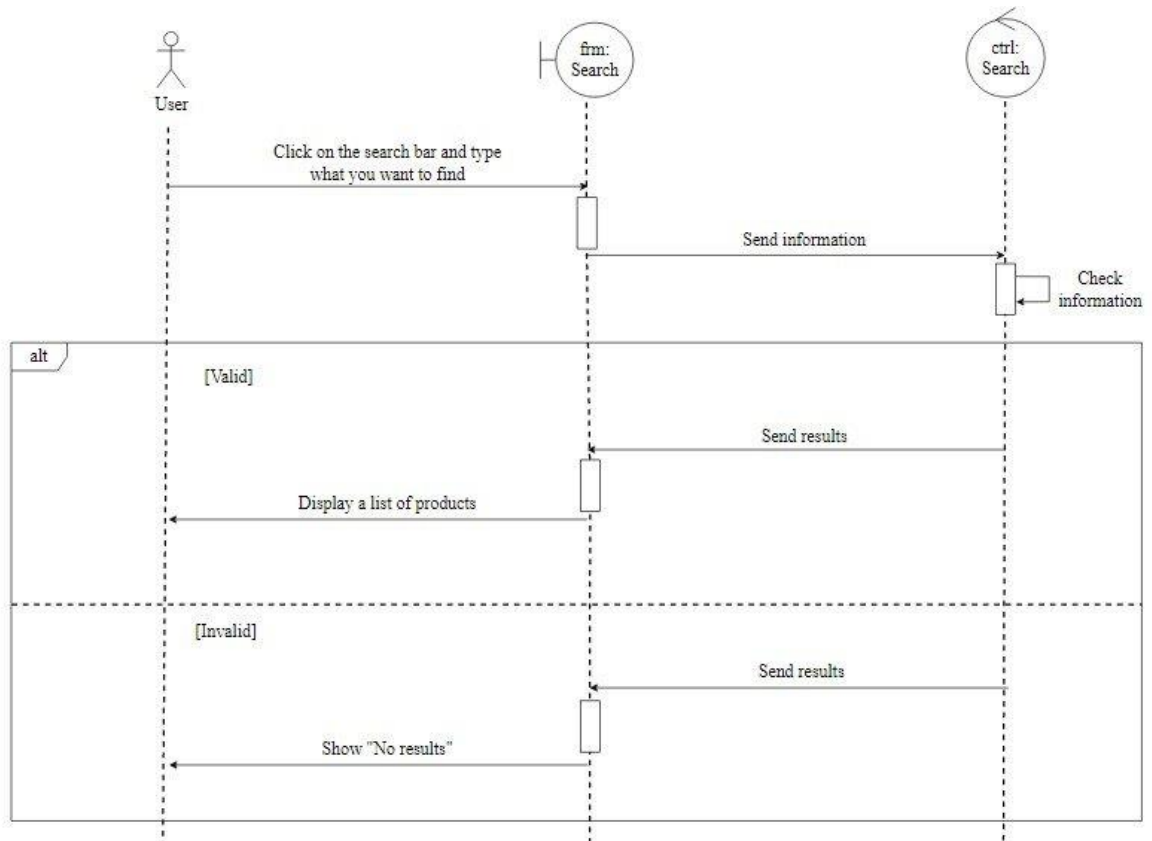
vi. Search for products

Image 28. Search for products sequence diagram

vii. View orders

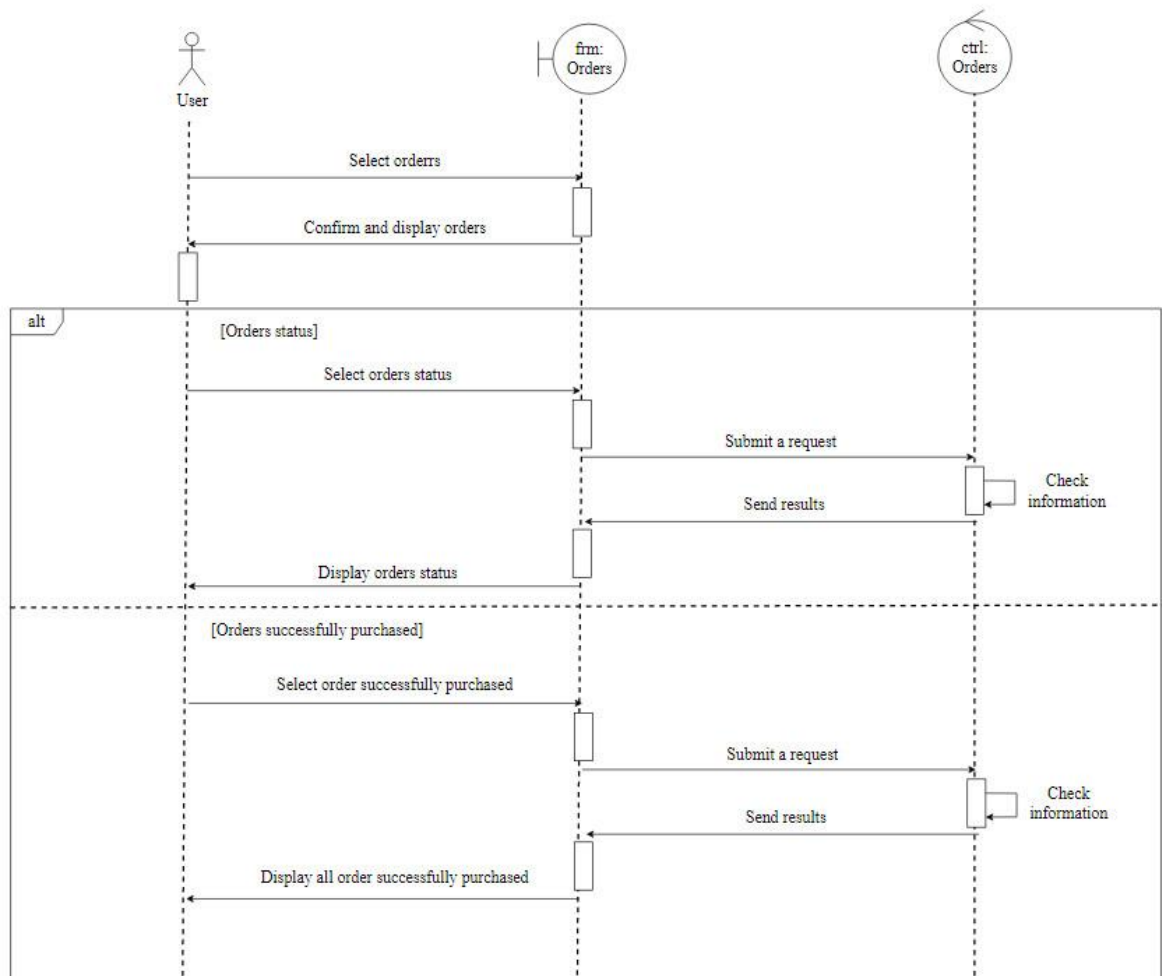


Image 29. View orders sequence diagram

viii. Customer support

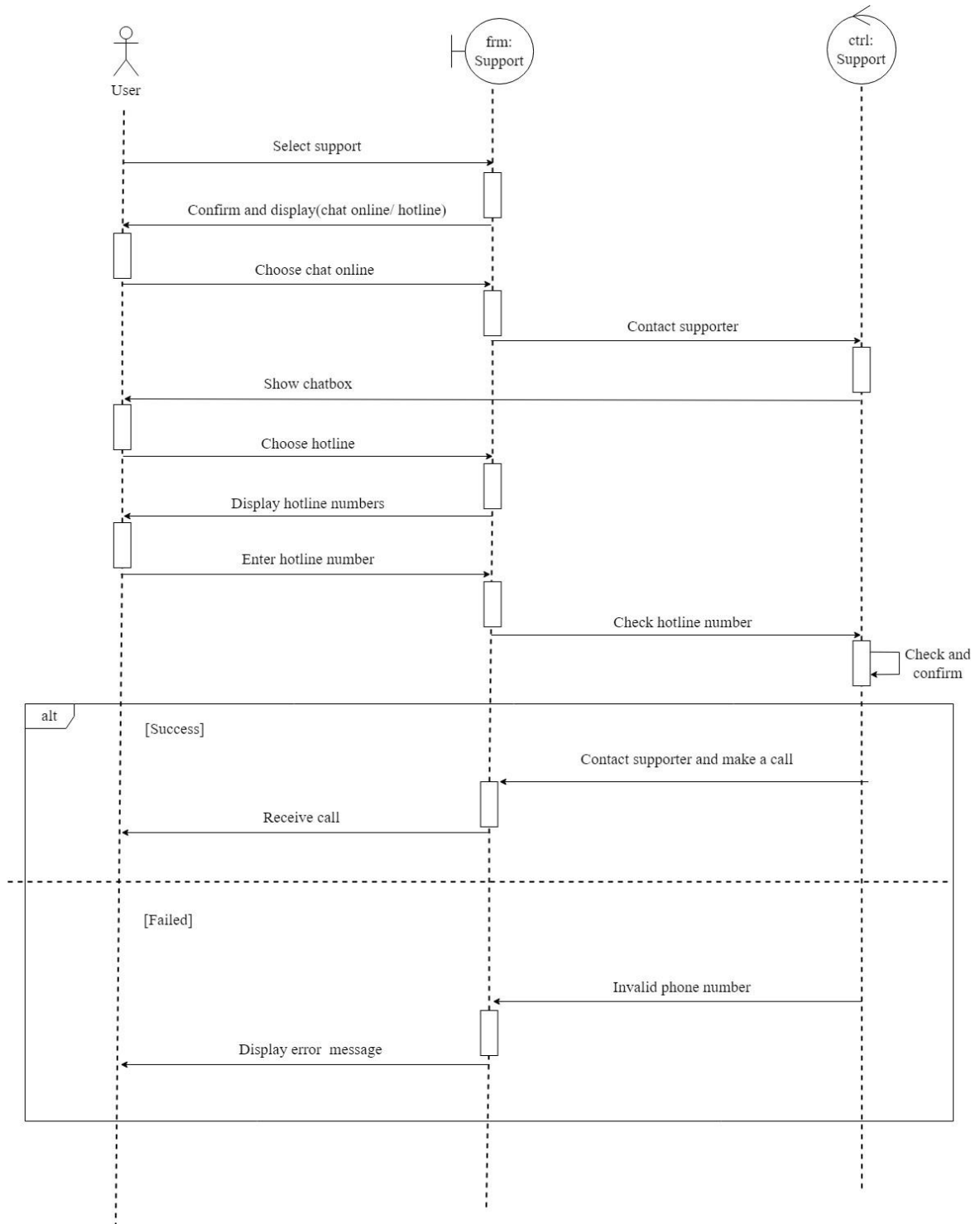


Image 30. Customer support sequence diagram

ix. Product portfolio management

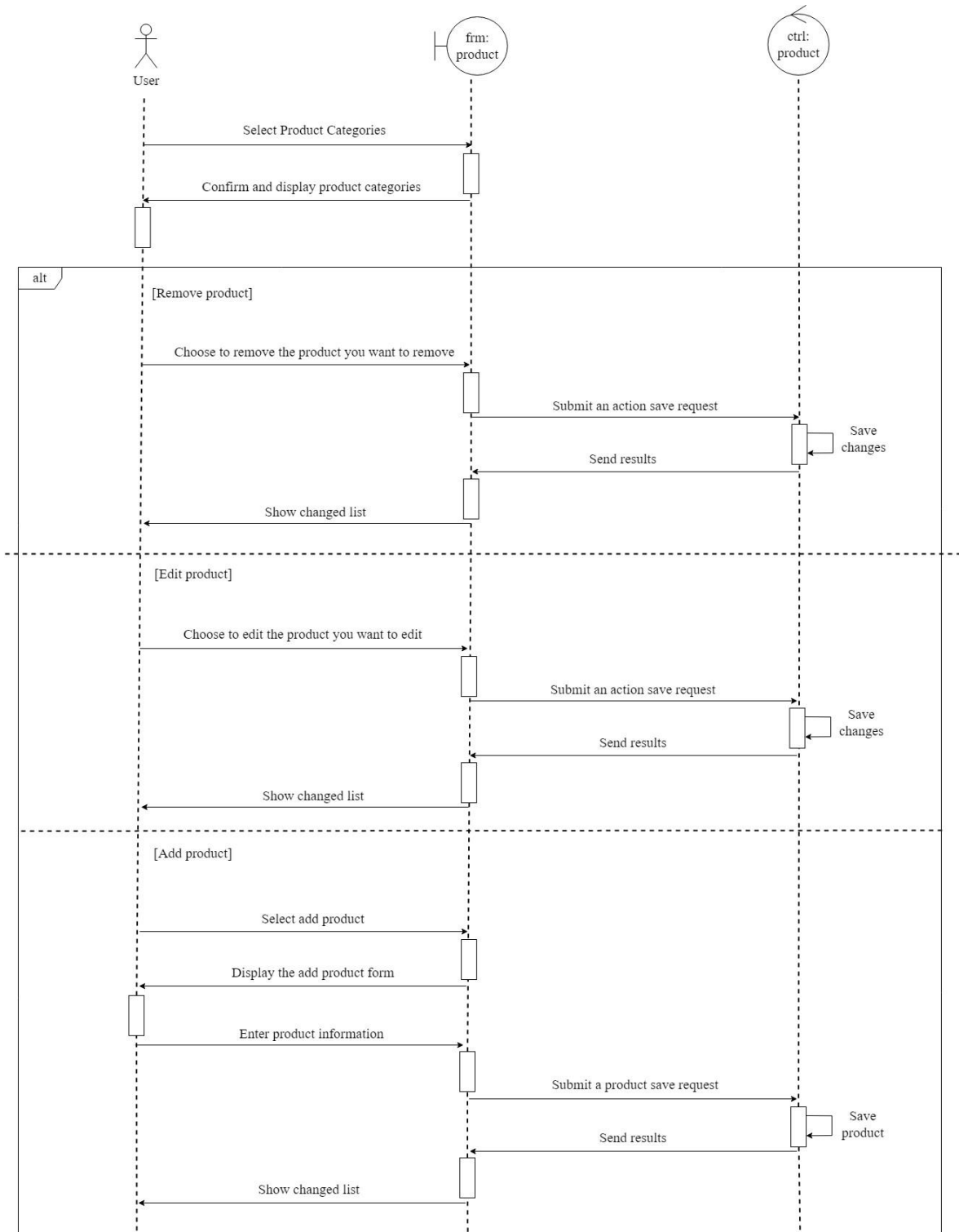


Image 31. Product portfolio management sequence diagram

x. Manage member lists

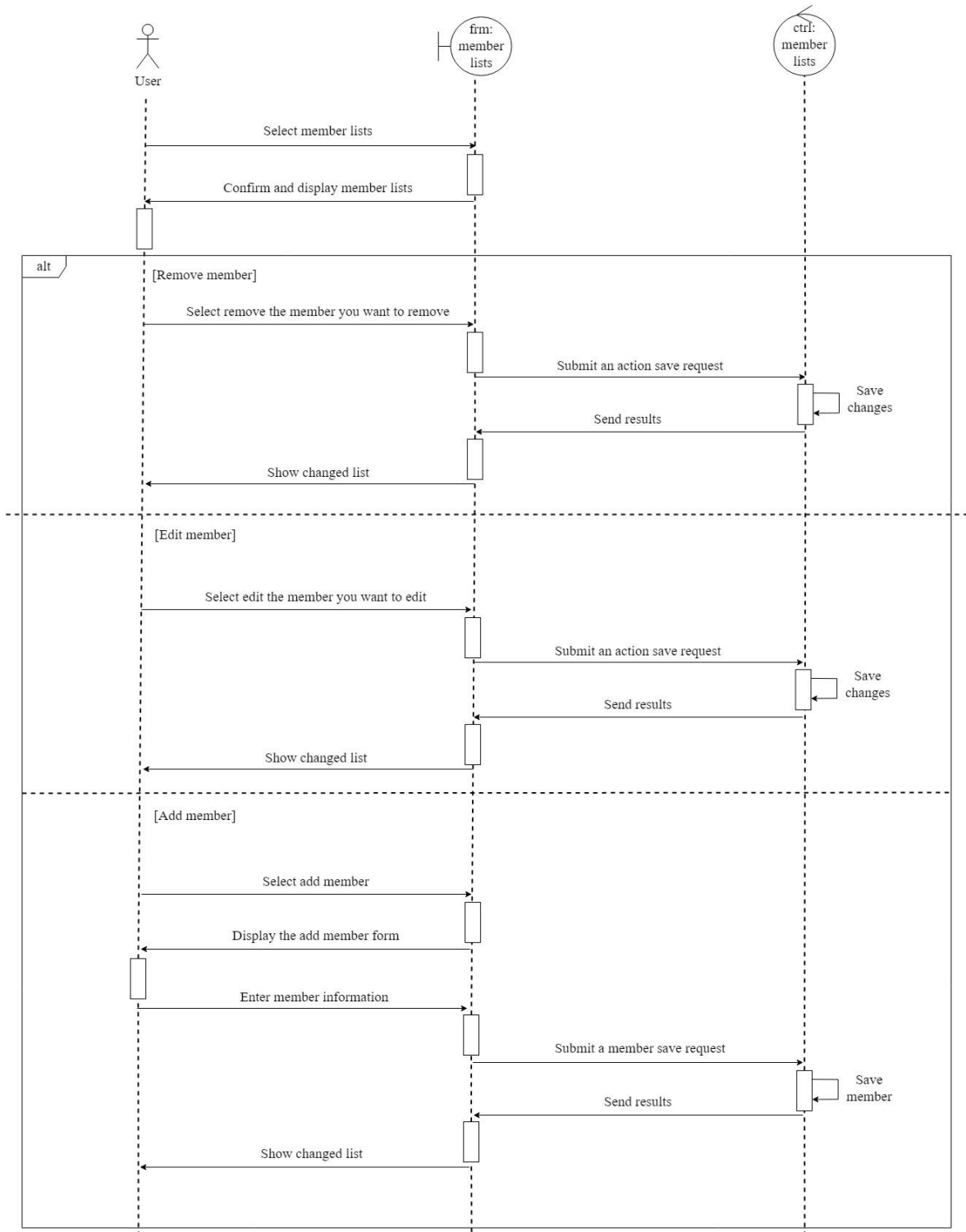


Image 32. Manage member lists sequence diagram

xi. Receiving and processing orders

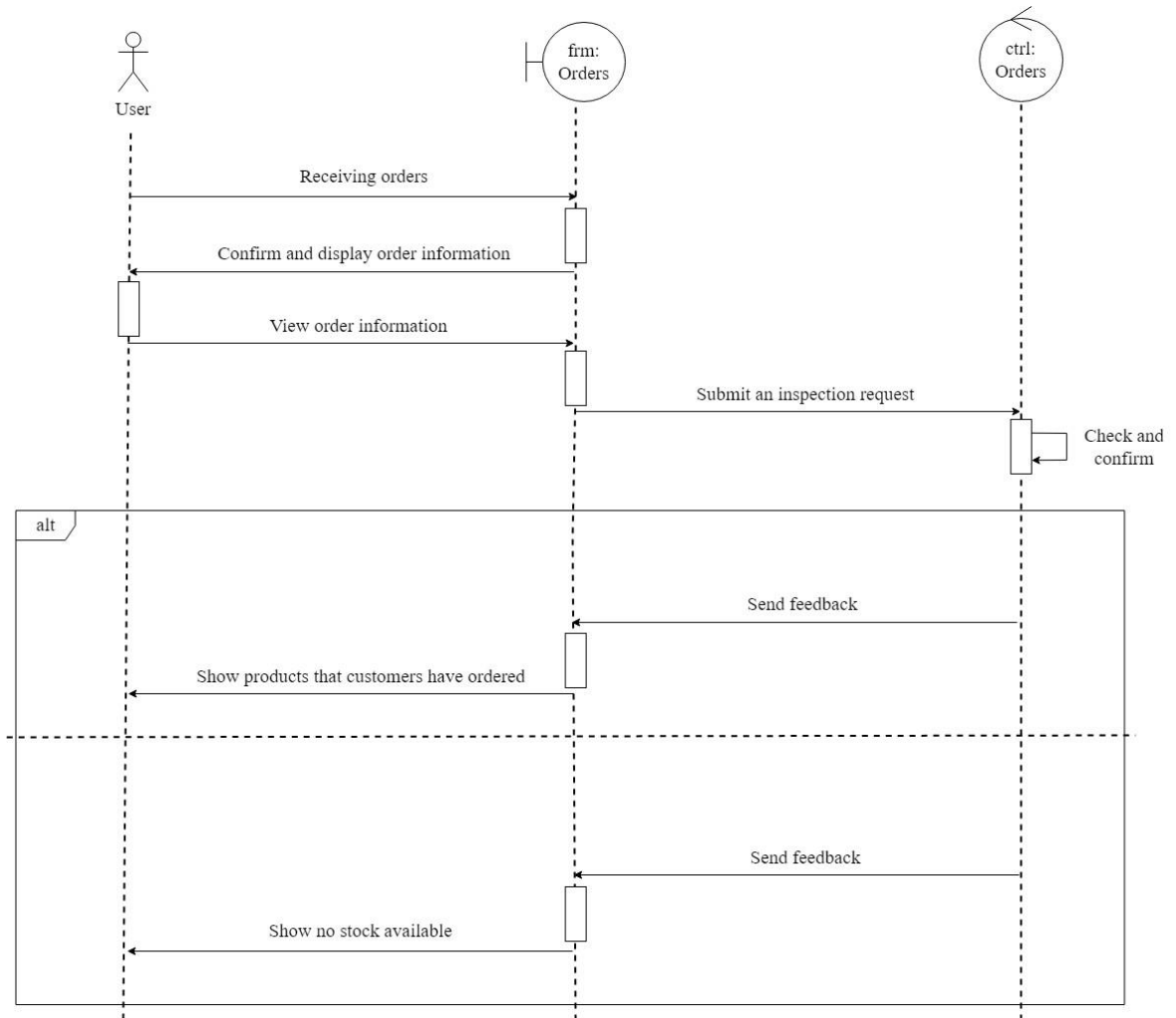


Image 33. Receiving and processing orders sequence diagram

5. Class diagram

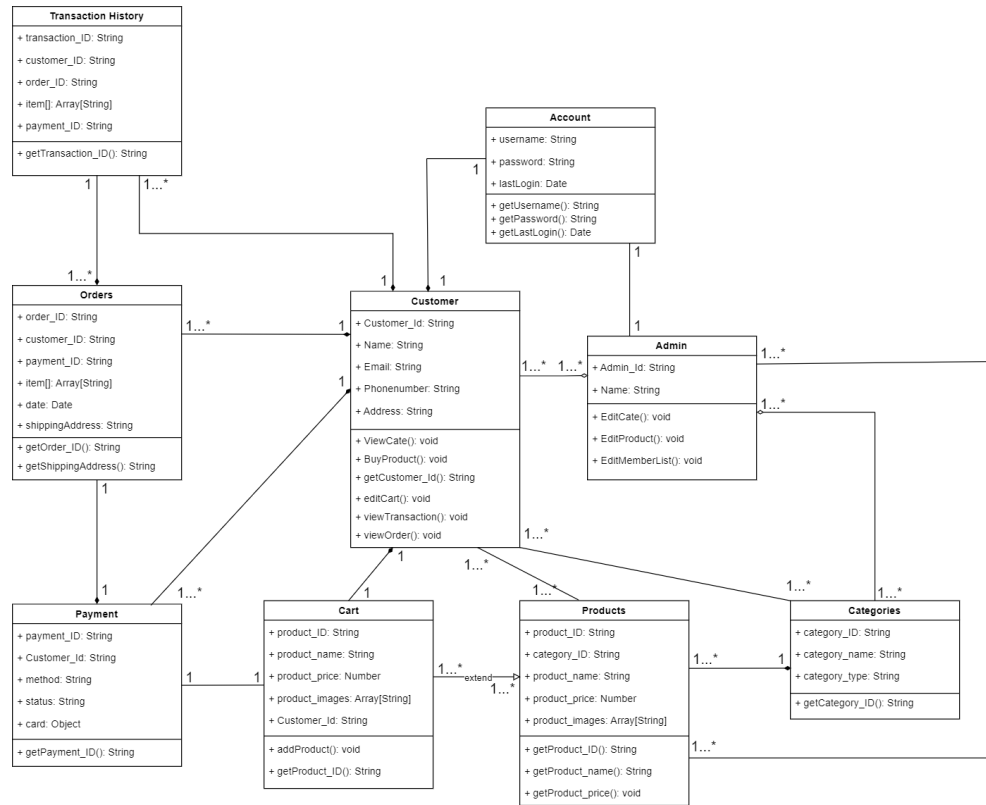


Image 34. Class diagram

6. Entity Relationship Diagram and Relational Data Model

i. Relation Data Model

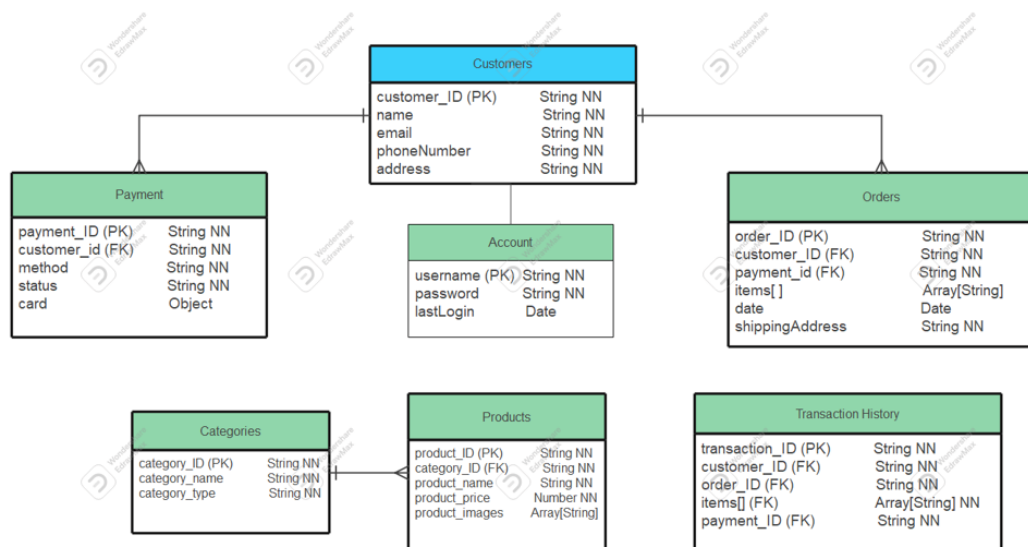


Image 35. Relation Data model

ii. Entity Relationship Diagram

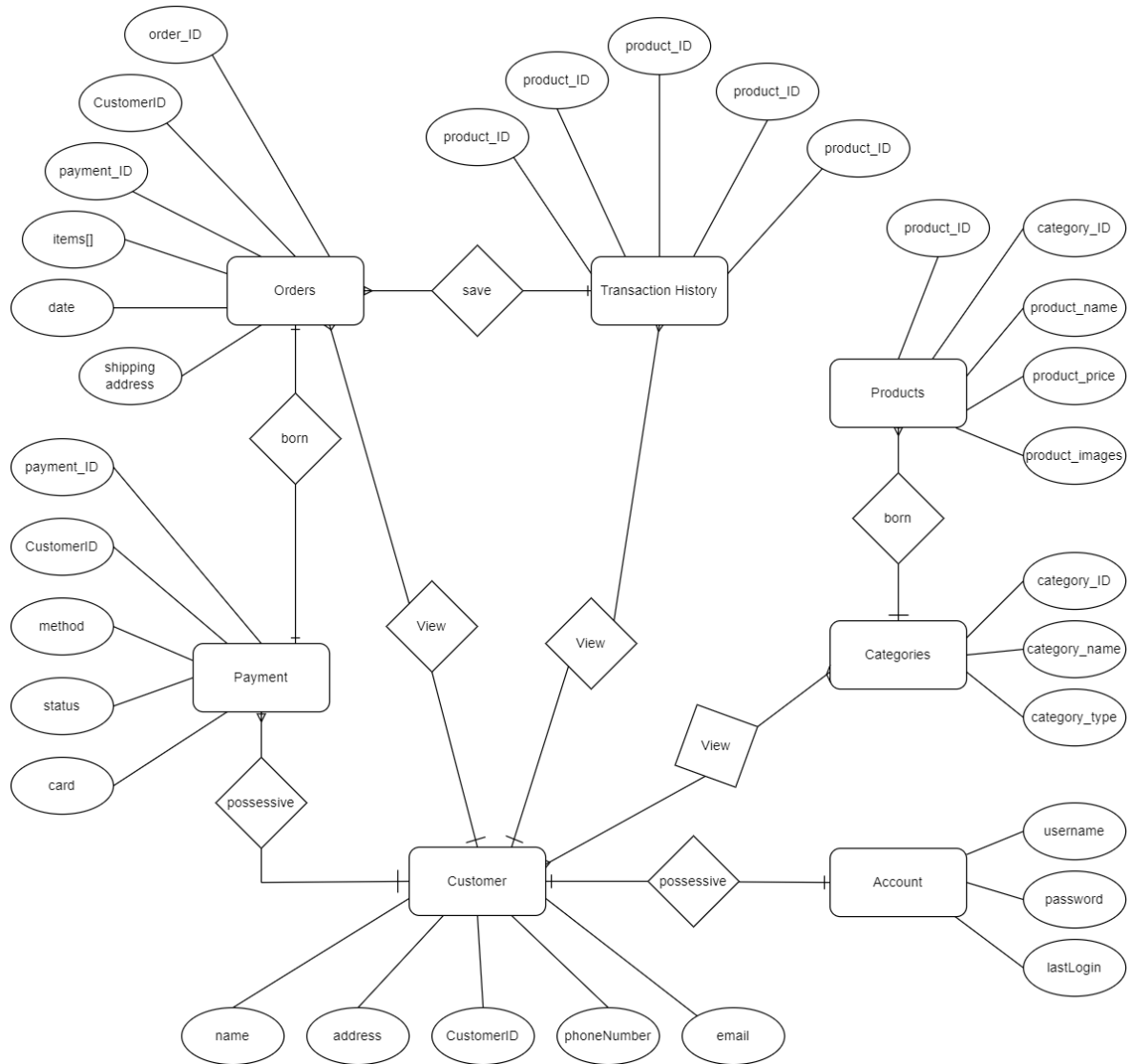


Image 36. Entity Relationship Diagram

CHAPTER IV: ACTUAL PROCESS

1. Software development life cycle model

i. Introduction to the waterfall model

The waterfall model is a sequential model that divides software development into predefined phases. The development process is like a flow, with phases performed in strict sequence and without rollbacks or phase jumps: analysis, design, build, test, implementation and maintenance.

The model was introduced by Winston Royce in 1970. The waterfall model is the earliest model in the Software Development Life Cycle.

Stages of development in the model:

- **Requirements Analysis:**
Stage of collecting and analyzing system requirements and documenting requirements specification
- **System Design:**
Software system analysis and design phase
Determine the overall architecture of the system
Discuss hardware and software requirements
- **Building the system (Implementation):**
The system is developed in units and integrated in the next phase.
Each unit (unit) developed and tested by programmer is called Unit Test.
- **System Testing:**
All units developed during the build phase are integrated into the system after unit testing has been completed. After integration, the whole system is checked and debugged.
- **System Deployment:**
After the testing steps are completed, the product is deployed to the customer and released to the market
- **System Maintenance:**
Maintain the system when there is any change from the client side.

ii. Advantages

- Is a simple model, easy to apply, sequential process step by step.
- Easy to manage because it is fixed in each step.
- Stages are explicitly processed and completed at the same time.
- Process and results are fully documented.
- Suitable for small projects with clearly defined requirements

iii. Defect

- The flexibility is not high, when a stage is in the process of being executed and an error is discovered, it will have to stop altogether and return to the first stage to update the document and start again from that stage..
- High risk, uncertain.
- Not suitable for projects with high complexity, many changes in requirements during the development life cycle.
- Difficult to measure development progress of each stage.

iv. Reason for choosing model

- Waterfall model is a simple model, easy to understand, easy to use.
- Suitable for disseminating and assigning work.
- The model has each specific phase, easy to apply and implement.
- The model is suitable for the implementation of the final project because the project has little change in requirements and the requirements are defined at the outset, fully and clearly.

2. Architecture use

Programming language used	Library use
Nodejs, CSS3, Handlebars	ExpressjsS

Table 16. Table of programming languages and libraries used

CHAPTER V: DEMO CODE

1. Customer interface

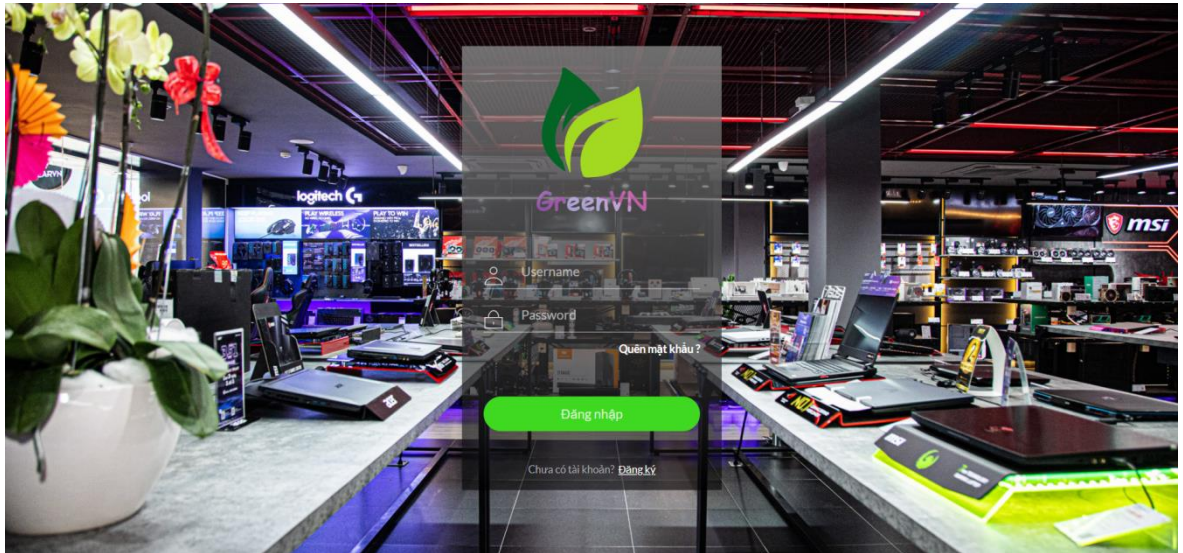


Image 37. Login screen

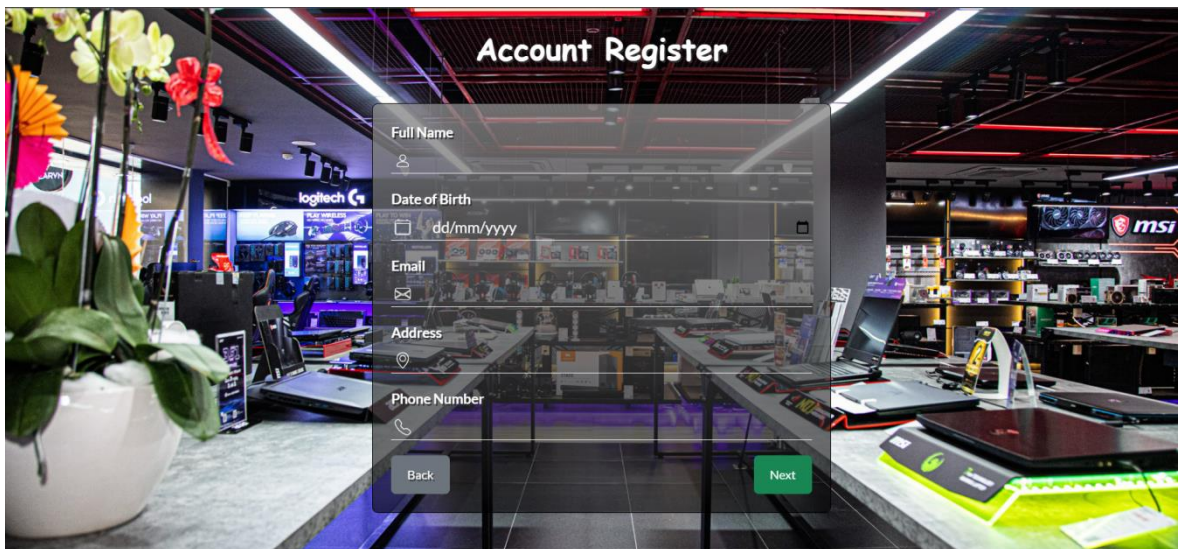


Image 38. Register screen 1

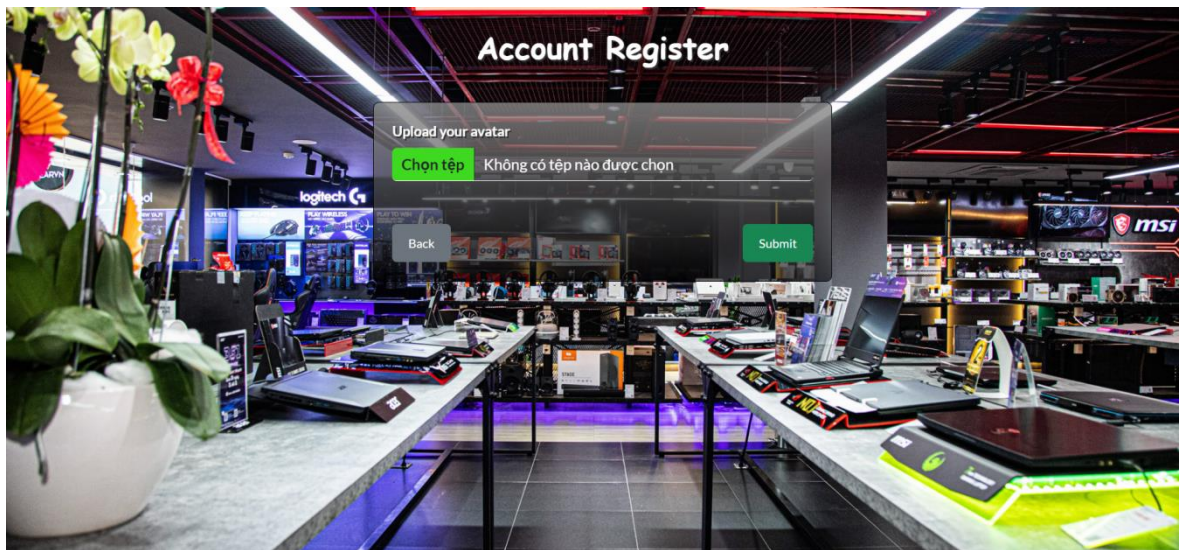


Image 39. Register screen 2

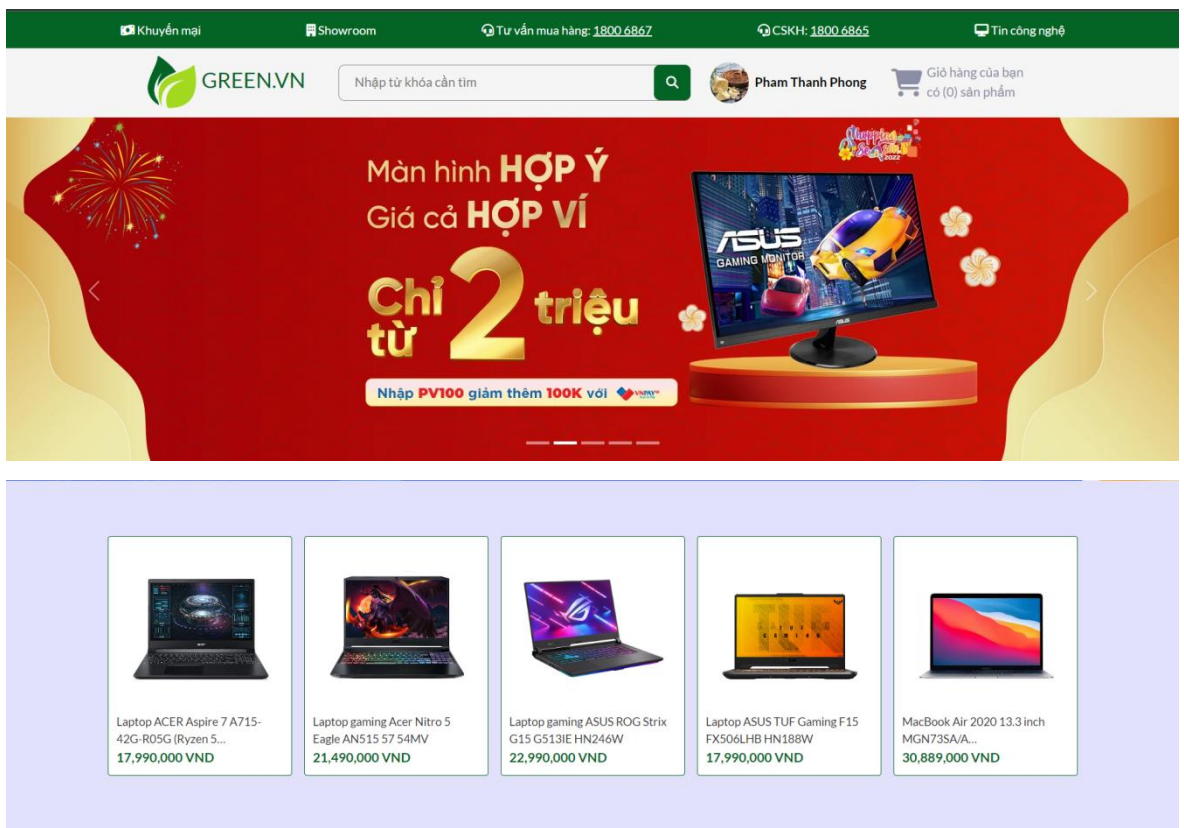


Image 40. Home screen



Laptop ASUS TUF Gaming F15 FX506LHB HN188W

- ✓ Bảo hành chính hãng 12 tháng.
- ✓ Hỗ trợ đổi mới trong 7 ngày.
- ✓ Windows bản quyền tích hợp.
- ✓ Miễn phí giao hàng toàn quốc.

ƯU ĐÃI KHI MUA KÈM LAPTOP TẠI GREENVN:

- 📦 Giảm ngay 100.000đ khi mua chuột TUF Gaming M3, Logitech G203, G102.
- 📦 Giảm ngay 100.000đ khi mua thêm màn hình máy tính.
- 📦 Giảm ngay 100.000đ khi mua thêm Ram.

Và còn rất nhiều ưu đãi khác. [XEM NGAY CHI TIẾT TẠI ĐÂY](#)

Giá: 17,990,000 VND

[Thêm vào giỏ hàng](#) [Đặt hàng](#)

Cấu hình chi tiết

Image 41. Product details screen 1

Cấu hình chi tiết

Cpu	Intel Core i5-10300H 2.5GHz up to 4.5GHz 8MB
Ram	8GB DDR4 2933MHz (2 khe ram, nâng cấp tối đa 32GB RAM)
HardDisk	512GB SSD M.2 PCIE G3X2, 1 khe SATA3 2.5
VGA	NVIDIA GeForce GTX 1650 4GB GDDR6 + Đồ họa Intel® UHD
Monitor	15.6 FHD (1920 x 1080) IPS, 144Hz Value IPS-level, 45% NTSC, 62.5% SRGB, Anti-glare display
Port	1x USB 2.0 Type-A, 1x USB 3.2 Gen 2 Type-C support DisplayPort™ / G-SYNC, 2x USB 3.2 Gen 1 Type-A, 1x HDMI 2.0b, 1x RJ45, 1x 3.5mm Combo Audio Jack
Audio	DTS:X® Ultra
Keyboard	Bàn phím Chiclet có đèn nền RGB
LAN	10/100/1000 Mbps
WIFI	802.11AX (2X2)
Bluetooth	v5.2 (Dual band) 2*2
Webcam	HD 720p CMOS module

Image 42. Product details screen 2

Hỗ trợ khách hàng

Thẻ ưu đãi

Trung tâm bảo hành

Thanh toán và giao hàng

Dịch vụ sửa chữa và bảo trì

Doanh nghiệp thân thiết

Chính sách Mua hàng & Bảo hành

Quy định chung

Chính sách Bảo mật Thông tin

Chính sách Vận chuyển và Lắp đặt

Chính sách bảo hành

Chính sách đổi trả và hoàn tiền

Quy định giá cả & hình thức thanh toán

Chính sách trả góp

Thông tin Green

Giới thiệu Green

Thông tin liên hệ

Hệ thống Showroom

Hội đáp

Tin công nghệ

Tuyển dụng

Cộng đồng Green

Gọi mua hàng 18006867

Gọi chăm sóc 18006865

Email liên hệ

Hỗ trợ Khách hàng: cskh@green.vn

Liên hệ báo giá: baogia@green.vn

Hợp tác phát triển: hoptac@green.vn

Phương thức thanh toán

QR Code

Tiền mặt

Trả góp

Internet Banking

Danh sách các ngân hàng thanh toán online



Image 43. Website info bar

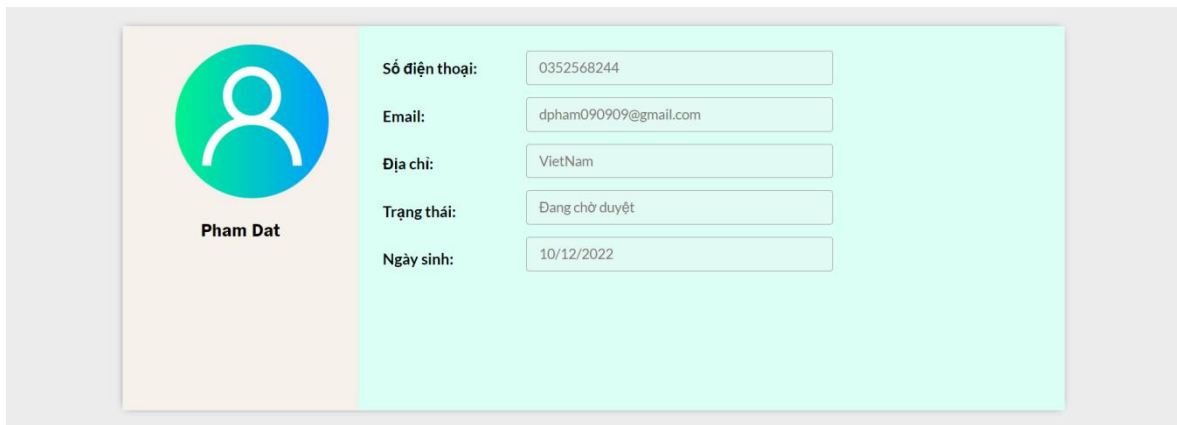


Image 44. Personal information screen

Giỏ hàng

☐ CÔNG TY CỔ PHẦN THƯƠNG MẠI DỊCH VỤ GREEN	Đơn giá	Số lượng	Thành tiền	Khuyến mãi
☐  Laptop ASUS TUF Gaming F15 FX506LHB HN188W	17,990,000 VND	- 1 +	17,990,000 VND	Thanh toán Tổng tạm tính 48,879,000 VND Thành tiền 48,879,000 VND Thanh toán Bạn cần đăng nhập để tiếp tục
☐  MacBook Air 2020 13.3 inch MGN73SA/A (M1/8GB/SSD512GB)	30,889,000 VND	- 1 +	30,889,000 VND	

Image 45. Cart screen

Thông tin người nhận hàng

Họ tên

Số điện thoại

Email

Địa chỉ nhận hàng

Tỉnh/Thành phố

Quận/Huyện

Phường/Xã

Địa chỉ cụ thể

Phương thức thanh toán

Thông tin thanh toán của bạn sẽ luôn được bảo mật

Thông tin đơn hàng



Laptop ASUS TUF Gaming F15 FX506LHB HN188W
 Số lượng **1**
 17,990,000 VND



MacBook Air 2020 13.3 inch MGN73SA/A (M1/8GB/SSD512GB)
 Số lượng **1**
 30,889,000 VND

Tổng tạm tính **48,879,000 VND**
 Phí vận chuyển **Miễn phí**
 Thành tiền **48,879,000 VND**

Image 46. Order screen

Phương thức thanh toán

Thông tin thanh toán của bạn sẽ luôn được bảo mật

Thanh toán bằng ví điện tử
 Momo, ZaloPay, ApplePay, VNPAY

Thanh toán khi nhận hàng

Thanh toán bằng thẻ ngân hàng
 Vietcombank, Agribank, ACB, NamAbank,...

Thanh toán bằng QR Code

Thanh toán

Image 47. Payment screen

Lịch sử mua sắm

ID	Date	Tổng bill	Phương thức	Các sản phẩm	Địa chỉ nhận hàng
o230425250	14/12/2022	17,990,000 VND	QR Code	 Laptop ACER Aspire 7 A715-42G-R05G (Ryzen 5 5500U/RAM 8GB/512GB SSD/ Windows 11) Số lượng 1 17,990,000 VND	Address: Duong835B, Phuoc Ly, Can Giuoc, Long An
o852545131	14/12/2022	30,889,000 VND	Tiền mặt	 MacBook Air 2020 13.3 inch MGN73SA/A (M1/8GB/SSD 512GB) Số lượng 1 30,889,000 VND	Address: Duong835B, Phuoc Ly, Can Giuoc, Long An
o926046927	14/12/2022	40,980,000 VND	Ví điện tử	 Laptop gaming ASUS ROG Strix G15 G513IE HN246W Số lượng 2 45,980,000 VND	Address: Duong835B, Phuoc Ly, Can Giuoc, Long An

Image 48. Shopping history screen

2. Admin interface

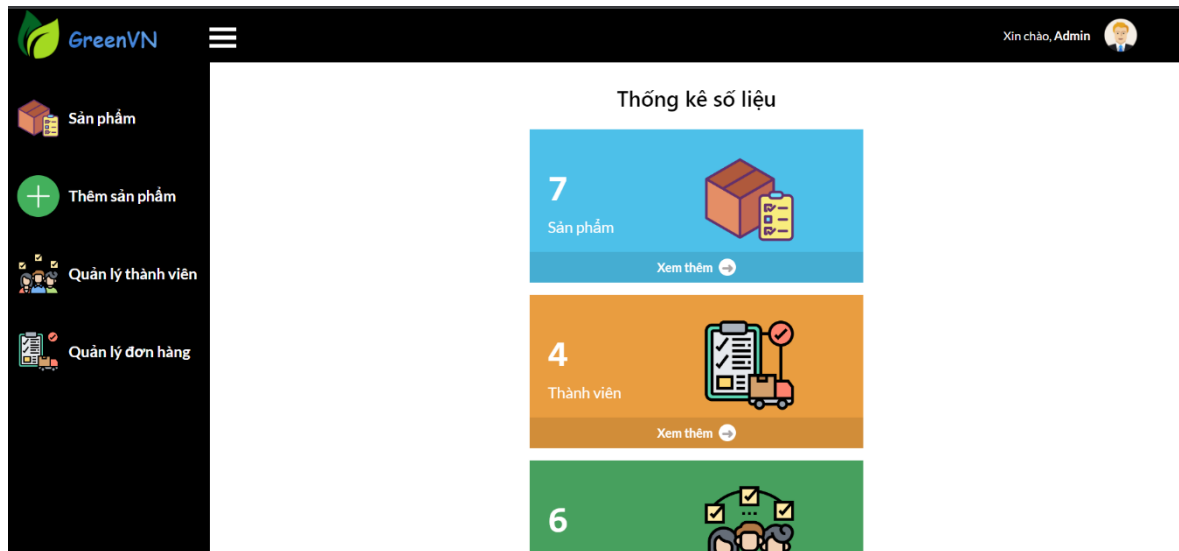


Image 49. Admin homepage

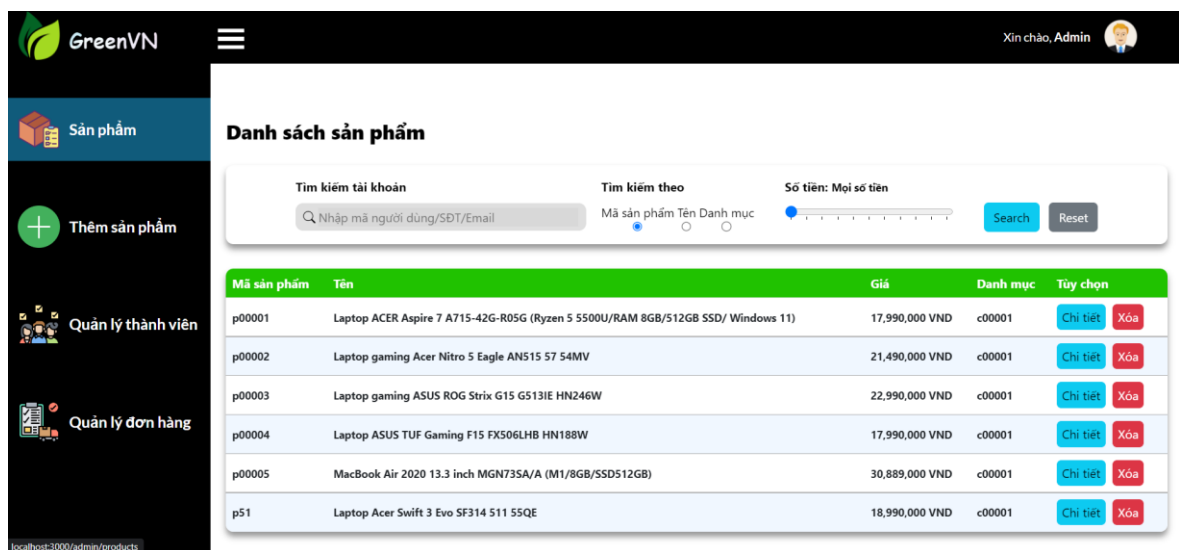


Image 50. Product management

GreenVN Xin chào,

Sản phẩm

Thêm sản phẩm

Quản lý thành viên

Quản lý đơn hàng

Danh sách yêu cầu rút tiền chờ được duyệt

Tìm kiếm giao dịch:
 Từ:
 Đến:
 Số tiền: Mọi số tiền

ID	Full Name	Phone Number	Email	Address
6544198410	Pham Thanh Phong	0398454023	thanhphong07022002@gmail.com	Some Where City
admin	Admin	0398454024	admin@gmail.com	Some Where City
0362354778	Pham Dat	0352568244	dpham090909@gmail.com	VietNam

localhost:3000/admin/usersManagement

Image 51. User management

GreenVN Xin chào,

Sản phẩm

Thêm sản phẩm

Quản lý thành viên

Quản lý đơn hàng

Danh sách yêu cầu rút tiền chờ được duyệt

Tìm kiếm giao dịch:
 Từ:
 Đến:
 Số tiền: Mọi số tiền

ID	Date	Tổng bill	Phương thức	Các sản phẩm	Địa chỉ nhận hàng
o230425250	14/12/2022	17,990,000 VND	QR Code	Laptop ACER Aspire 7 A715-42G-R05G (Ryzen 5 5500U/RAM 8GB/512GB SSD/ Windows 11) Số lượng 1 17,990,000 VND	Address: Duong835B, Phuoc Ly, Can Giuoc, Long An
o852545131	14/12/2022	30,889,000 VND	Tiền mặt	MacBook Air 2020 13.3 inch MGN735A/A (M1/8GB/SSD512GB) Số lượng 1 30,889,000 VND	Address: Duong835B, Phuoc Ly, Can Giuoc, Long An
o926046927				Laptop gaming ASUS ROG Strix G15 G513IE	

localhost:3000/admin/ordersManagement

Image 52. Order management

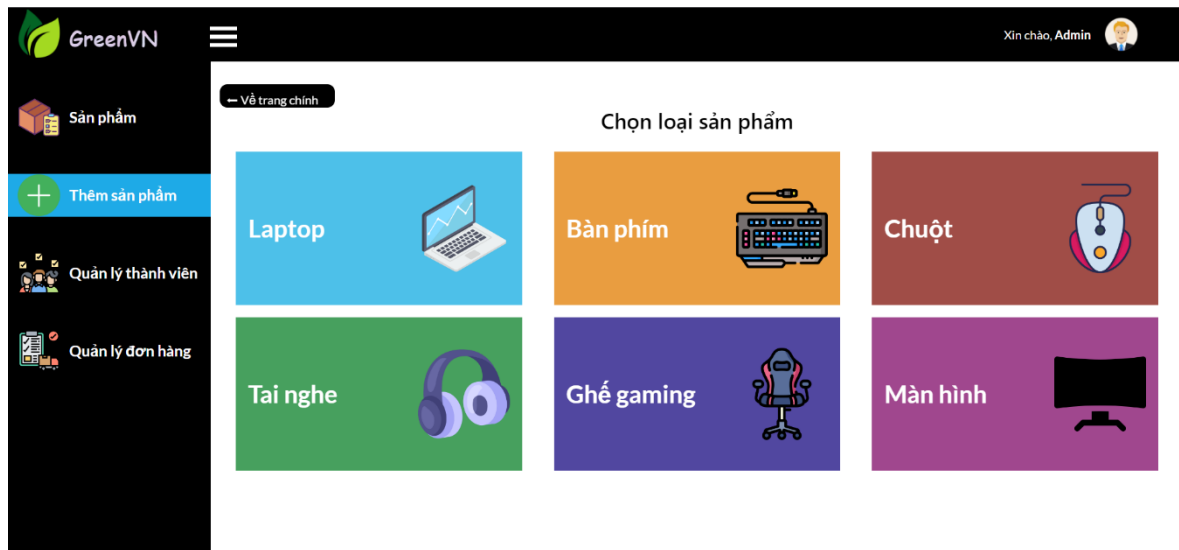


Image 53. Add category page

CHAPTER VI: TEST CASE AND UNIT TEST

ID	Unit test	Test case	Result
1	Owner login	Enter “admin” for username and “1234567” for password	Fail
2	Owner login	Enter “admin” for username and “123456” for password	Success
3	Adding category	Missing any information	Fail
4	Adding category	Wrong data type	Fail
5	Adding category	Duplicate primary key (cateID)	Fail
6	Adding category	Fill all blanks and correct data types	Success
7	Updating quantity of product when print bill	Sold quantities > stock quantities	Fail
8	Updating quantity of product when print bill	Sold quantities < stock quantities	Success

9	Total amount of a bill	Sold total > Total of bill	Fail
10	Total of transaction	Greater than the number of incomes	Fail
11	Best seller	Compare the total sale bases on staff's name	Success

Table 17. Test Case and Unit Test

REFERENCES

- [Link 1](#)
- [Link 2](#)
- [Link 3](#)
- [Link 4](#)
- [Link 5](#)